A Major Project Phase-II Report

On

# "DESIGN AND IMPLEMENTATION OF CIRCULAR POLARIZED PATCH ANTENNA ARRAY USING HIGH IMPEDANCE SURFACE"

SUBMITTED IN PARTIAL FULFILLMENT OF THE

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**BACHELOR OF ENGINEERING**

**IN**

**ELECTRONICS & COMMUNICATION ENGINEERING**

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# Certificate

This is to certify that the Major Project Phase-II report entitled **"DESIGN AND IMPLEMENTATION OF CIRCULAR POLARIZED PATCH ANTENNA ARRAY USING HIGH IMPEDANCE SURFACE"** being submitted by **Ms. KHATIJA MAHVEEN** (1608-20-735- 084), **Ms. M ALEKHYA** (1608-20-735-090) and **Mr. B PRANAY KUMAR**(1608-20-735-319) in partial fulfillment for the award of the Degree of Bachelor of Engineering in Electronics and Communication Engineering of the Osmania University, Hyderabad, during 2023-24, is a record of bonafide work carried out under our guidance and supervision.

The results presented in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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# DECLARATION

This is to declare that the work submitted in the present project work report titled **"DESIGN AND IMPLEMENTATION OF CIRCULAR POLARIZED PATCH ANTENNA ARRAY USING HIGH IMPEDANCE SURFACE"** is a record of bonafide work done by me in the Department of Electronics & Communication Engineering, Matrusri Engineering College, Saidabad, Hyderabad.

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We would like to take this opportunity to place it on the record, that this project would never have taken shape but for the cooperation extended to us by certain individuals. Though this is not possible to name all of them, it would be a pardonable on us part if we don't mention some of the very important persons. Sincerely we acknowledge our deep sense of gratitude to the project guide, **Dr. K. KOTESWARA RAO**, Assistant Professor for his constant encouragement, help and valuable suggestions. We wish to thank him for his constant motivation and help throughout the project.

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# ABSTRACT

The project focuses on the design and implementation of a circular polarized patch antenna array utilizing a high-impedance substrate. Circular polarization enhances robust signal reception. The use of a high-impedance substrate contributes to miniaturization and improved performance of the antenna array.

The antenna geometry and feed mechanism are designed to achieve circular polarization and efficient impedance matching. HFSS is employed to model and optimize the antenna array. The simulation results guide iterative refinements to the design, ensuring the array meets predefined performance criteria. The fabrication process involves precision techniques like photo-lithography and substrate layering, to maintain accurate dimensions and substrate thickness. Prototypes are fabricated and tested using network analyzers.

The outcomes of this project contribute to advancements in antenna design, particularly in the context of circular polarization and high-impedance substrates. Applications of the circular polarized patch antenna array make it suitable for various communication systems, satellite applications, and wireless networks.

**Keywords:** Circular polarization, miniaturization, multipath propagation, antenna design, electromagnetic simulation, patch antenna array, prototyping.

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# LIST OF ABBREVIATIONS

| | |
|---|---|
| PAN | Personal Area Network |
| WLAN | Wireless Local Area Network |
| GHz | Giga Hertz |
| MHz | Mega Hertz |
| PBG | Photonic Band Gap |
| EBG | Electromagnetic Band Gap |
| FR-4 | Flame Retardant Class-4 |
| HFSS | High Frequency Structure Simulator |
| CST | Computer Simulator Tool |
| EM | Electromagnetic |
| BW | Bandwidth |
| TEM | Transverse Electromagnetic |
| SMA | Sub-Miniaturized Version A |
| CP | Circular Polarized |
| LP | Linear Polarized |
| TE | Transverse Electric |
| TM | Transverse Magnetic |
| 2D | Two Dimensional |
| 3D | Three Dimensional |
| ISM | Industrial Scientific and Medical |
| MSA | Microstrip Antenna |
| ADS | Advanced Design System |
| PMC | Perfect Magnetic Conductor |
| VSWR | Voltage Standing Wave Ratio |
| AR | Axial Ratio |
| RF | Radio Frequency |
| FEM | Finite Element Method |
| VNA | Vector Network Analyzer |

# Chapter 1

# INTRODUCTION

## 1.1 Introduction

An antenna is a specialized transducer that converts electric current into electromagnetic (EM) waves or vice versa. Antennas are used to transmit and receive non ionizing EM fields, which include radio waves, microwaves, infrared radiation (IR) and visible light. An antenna is a device or mechanism that is made of metallic material and absorbs or emits electromagnetic waves, also called electromagnetic radiation. Antennas are used for many types of telecommunication, a type of long-distance communication that uses radio waves to transmit messages which are then converted into audio or other mediums.

Antennas are often categorized as either transmitting or receiving. However, many antennas can do both through a transceiver. A transmitting antenna receives current from a transmitting device. From this current, the antenna generates EM waves at a specific frequency that radiate out through the air, where they can then be received by one or more other antennas.Antennas are designed to transmit and receive EM waves at specific frequencies. Their design is also determined by such factors as direction, movement and signal strength. This is why antennas come in so many different shapes and sizes. For example, car antennas are designed much differently from television antennas, and both types of antennas are designed much differently from microwave antennas and cell phone antennas.

There are several different types of antenna, and each has a specific purpose. For example, AM, FM, and Wi-Fi antennas all transmit different wavelengths. AM antennas are long because they transmit long wavelengths, while Wi-Fi router antennas are much shorter because they transmit much shorter wavelengths. In addition, parabolic antennas can focus or amplify signals they send and receive.

**Types of Antennas:**

1. Reflector Antenna : An antenna that includes one or more components that reflect the EM waves in order to better focus or direct them. Reflector antennas are often used in microwave and satellite communications. Many include a parabolic structure that reflects EM waves, such as those used in satellite dishes.

2. Lens Antenna : An antenna that includes an embedded lens made up of glass, metal or a dielectric material. The antenna uses the convergence and divergence properties of the lens to transmit or receive EM waves, typically at higher frequencies. Lens antennas are often used for radar systems and microwave communications.

3. Log periodic Antenna : A directional antenna with multiple elements that can support a broad range of frequencies. The supported range depends on the size of the elements and how they're arranged, which is based on a logarithmic function of frequency. Log periodic antennas can be useful in situations that require variable bandwidth or that support high-frequency communications, such as analog televisions, cellular communications or shortwave radios.

4. Microstrip Antenna : A small antenna printed into a circuit board. The antenna itself is a patch made out of a conductive material that is mounted on a dielectric substrate, which itself sits on a ground plate. Microstrip antennas are used extensively in wireless communication and mobile devices including cell phones.

5. Traveling-wave Antenna : A directional antenna in which the EM waves travel through the antenna in one direction, unlike many other types of antennas in which the waves travel in multiple directions. The unidirectional waves make it possible to support a wider range of frequencies. Traveling-wave antennas are used for analog televisions, amateur radios, telecommunications and other use cases.

6. Wire Antenna : An antenna that is nothing more than a length of wire, connected at one end to a transmitter or receiver. Wire antennas are the simplest and most portable type of antenna. They're used extensively with radios, automobiles, ships, aircraft, buildings and a variety of other devices and structures.

**Microstrip patch antenna** are a class of printed type of antenna, consisting of a substrate sandwiched between a ground plane and a patch. They are fabricated by etching out a piece of patch of conductive material on top of a dielectric substrate that is mounted on a ground plane made of conducting material as that of patch. Excitation to the antenna is provided using feed-lines connected to the patch. Microstrip patch antennas offer several advantages over conventional microwave antennas, covering the broad frequency range from ~100 MHz to ~100 GHz.Microstrip patch antenna provides attractive solution to compact, portable and affordable range of wireless products. They are becoming increasing useful as they can be directly printed onto the circuit board.

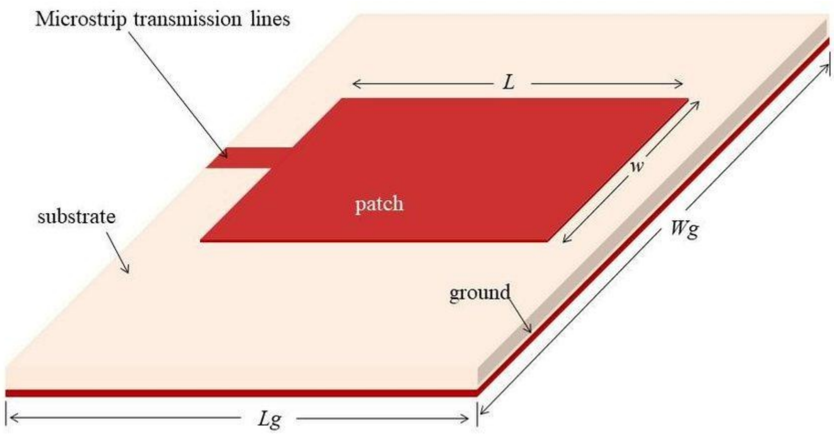


**Figure 1.1: Rectangular Microstrip Patch Antenna**

They introduce geometric perturbations that generate the necessary modes of excitation required for the respective polarization. Feed lines and matching networks can be fabricated simultaneously with the antenna structure that allows for proper impedance matching to reduce reflections and improve antenna efficiency.

However, they suffer from their own set of limitations and drawbacks. They deliver narrow bandwidth with tolerance problems, marginally lower gain (~6dB), ohmic losses and spurious fields from the feed structure. They have low power handling capacity and exhibit reduced antenna gain and efficiency. It suffers from high levels of cross-polarization and mutual coupling at high frequencies.

In patch antenna arrays, mutual coupling causes excitation of the adjacent patches by the energy radiated from the target patch, thereby deteriorating the overall radiation pattern. An attempt to reduce the size of printed patch antenna severely degrades antenna performance and results in increased side lobe activity. This implies a fundamental limit on patch antenna miniaturization. Another phenomenon called Surface Wave Excitation in patch antennas drastically lowers the antenna gain and efficiency, causing poor end-fire radiation.

## 1.2 Problem Specification

A patch antenna has several limitations like low gain, narrow bandwidth so here in our project we are making use of an array. However, using an antenna array causes problems like being easily affected by interference, surface wave propagation issues, and having a narrow bandwidth. To solve these issues, we are using Electromagnetic Bandgap Structures (EBG) structures, which help to overcome these drawbacks and improve the antenna characteristics.

## 1.3 Objectives

1. To design, simulate and fabricate microstrip patch antenna array for ISM band application at 2.4 GHz.
2. Implement EBG structure to improve antenna performance.
3. Employ circular polarization to overcome communication link drawbacks.

## 1.4 Specifications

The following are the desired specification of Microstrip Patch Antenna Array design:

**Table 1.1: Antenna Hardware Specifications**

| S.No. | Parameters | Value |
|---|---|---|
| 1. | Frequency of Operation | 2.4 GHz |
| 2. | Substrate Material | FR-4 |
| 3. | Substrate Permittivity | 4.4 |
| 4. | Substrate Height | 1.6 mm |
| 5. | Patch and Ground Material | Copper |

| | | |
|---|---|---|
| **6.** | Feeding Technique | Inset feeding<br>CPW |
| **7.** | Circular Polarization Technique | Truncated Edges |

## 1.5 Methodologies

Beginning with a thorough literature review, existing research on circular polarized patch antennas and high impedance substrates is examined. Design specifications are then established, outlining the desired parameters such as frequency, gain, and performance criteria. The selection of a suitable high impedance substrate follows, taking into account material properties and impedance matching considerations. Individual circular polarized patch antennas are designed using simulation tools, with optimization focused on achieving desired performance.

Subsequently, the configuration of the antenna array is determined, considering factors like array type and optimized spacing between antennas. The designed array undergoes simulation and analysis using electromagnetic simulation software to assess circular polarization, impedance matching, and overall performance metrics. With the finalized design, a prototype of the circular polarized patch antenna array is fabricated, adhering to manufacturing constraints and material properties. Practical tests are conducted to validate the performance of the fabricated array, comparing experimental results with simulated outcomes.

The design is iterated upon based on testing outcomes, optimizing for enhanced performance. The entire design and implementation process is meticulously documented, encompassing challenges faced and solutions devised. The project concludes with the preparation of a comprehensive report detailing the methodology and outcomes.

## 1.6 Motivation

This project is driven by the need for improved communication systems and the desire to advance antenna technologies. Circular polarized patch antenna arrays are a key focus, offering benefits like reduced signal distortion and enhanced performance. The inclusion of a high impedance substrate adds innovation to the project, potentially improving overall efficiency.

The motivation is rooted in addressing the demand for better communication capabilities, especially in modern systems requiring reliable signal transmission.The versatility of circular polarized patch antenna arrays, applicable in various communication scenarios, underscores the project's aim to create an adaptable solution. From a research perspective, the project contributes to existing knowledge, potentially influencing future developments in antenna technology.

## 1.7 Applications

1. ISM band applications- Bluetooth (PAN), Wi-Fi (WLAN)
2. Amateur Satellite Radio Communications
3. Cellphone Antennas
4. Future 5G-6G Communication Systems
5. Phased Array Radar Systems
6. Military Applications

## 1.8 Project Layout

The overall thesis is organized into nine chapters.

Chapter 1: Provides a brief introduction about the microstrip patch antenna array along with the objectives of the thesis.

Chapter 2: Provides the literature review and the theory involved in the research work.

Chapter 3: Literature on patch antenna, their parameters, types of polarization, feeding techniques, circular polarization techniques.All the antenna parameters included in this thesis are surveyed.

Chapter 4:Investigates the proposed system design of EBG surface and use of it .

Chapter 5:Investigates the proposed system design of circular polarized patch antenna array with EBG surface, along with the antenna equations required to design a simple patch antenna.

Chapter 6:Provides a brief description on the simulation software to be used. Design methodology along with simulation steps are presented. Simulated antenna designs along with EBG unit cell have been presented here..

Chapter 7:Presents the hardware implementation of circular polarized patch antenna array with EBG surface. Fabrication steps along with actual fabricated hardware and their results as measured in vector network analyzer are presented.

Chapter 8: Showcases both simulated and hardware results. Simulation results are compared to identify the enhancements due to EBG surface implementation. Also, simulated and hardware results are compared to ensure validity of the antenna design.

Chapter 9:Summarizes all the important observations and results from previous chapters.Potential future work is also put forth.

# Chapter 2

# LITERATURE SURVEY

**1. O. Ayop and M. K. A Rahim, "Analysis of mushroom-like electromagnetic band gap structure using suspended transmission line technique", IEEE International RF & Microwave Conference, pp. 258-261, 2011.**

In this report, an analysis of a mushroom EBG (EBG) structure using method of suspended transmission line has been discussed. This structure consists of nine elements of metallic patch grounded with metal. The parametric studies are patch width, substrate thickness, relative permittivity, gap width between adjacent patch and the size of via radius. It involves the study of band-stop and band-pass characteristic of the electromagnetic bad gap structure. The result shows that the band-stop frequency of EBG structure is affected when varying each of the parameters. This is due to the changes of the equivalent LC network of the mushroom-like EBG structures. By increasing the value of the patch width and relative permittivity, the operating frequency and bandwidth will decay due to the increment of the total capacitance value. Varying the gap between adjacent patches does not give significant effect to the band gap frequency because it only changes the value of fringe capacitor, which is very small compared to the overall capacitances. Increasing the size of via radius will increase the frequency band gap and bandwidth due to the reduction of total capacitance. Lastly, increasing the substrate thickness will reduce the operating frequency however increase the bandwidth. This is because the value of inductor increases more than the increment of the total capacitance. These relations prove to be useful in parametric design.

**2. B. Ramesh, Dr. V. Rajya Lakshmi, "Design Of A Rectangular Microstrip Antenna Using EBG Structure", International Journal of Engineering Research & Technology (IJERT) Volume 02, Issue 07, July 2013.**

The paper presents a work having a design of circularly polarized microstrip antenna with dual feed and employs Electromagnetic Band Gap (EBG) structure to improve antenna efficiency and bandwidth. A cross slot is introduced with equal arm lengths. The antenna is designed to work on Taconic RF-30 having thickness of 0.76mm and $\epsilon r$ of 2.92.The proposed EBG structure constructed with many vacuum holes. The designed antenna is simulated using CST Microwave Studio Version 12. Without EBG, resonant at 5.5GHz return loss of -16.9dB was observed having 410MHz bandwidth. With EBG at 5.5GHz, return loss of -17.47dB and 460MHz bandwidth was evaluated, at -10dB criteria. Improvement in directivity of 0.2dBi and 50 MHz bandwidth was realized on the antenna structure implementing the vacuum EBG structure.

**3. S. Swathi and S. Murugan, "EBG based Circularly Polarised Microstrip Antenna for RF applications", International Conference on Communications and Signal Processing (ICCSP), pp. 1786-1790, 2015.**

This paper studies the proposed design of a monopole capacitive feed square ring circularly polarized microstrip patch antenna covering 2.3-2.7 GHz frequency range, for ISM band applications. The geometry incorporates L-shaped and lightening-shaped strips for achieving circular polarization and capacitive feed strip fed by a coaxial probe as feeding technique. A 1-dimensional electromagnetic band gap (EBG) structure is incorporated without vias in the design to reduce the surface waves, to increase the gain, axial ratio and bandwidth. Parameters such as return loss (Sl1), voltage standing wave ratio (VSWR), impedance, axial and radiation pattern are simulated to analyze the antenna performance. The simulation showed that the impedance bandwidth of 1.54% and axial ratio bandwidth of 0.77% is obtained at 2.47GHz center frequency. The truncated square ring capacitive feed MSA with partially monopole ground plane offers better impedance bandwidth in the frequency range 2.453GHz-2.490GHz, which includes the desired frequencies of ISM. Axial Ratio is improved by using L-strips, lightning strips and varying truncation. Return loss improved by 10.5dB and impedance bandwidth by 11.5MHz for 2.4GHz resonance, for patch antenna with EBG structure. Hence, overall improvement is observed.

**4. Vinod Dohre, R. K. Prasad, “A Survey on Feeding Techniques of Microstrip Patch Antenna”, IJSRD - International Journal for Scientific Research & Development, Volume 3, Issue 06, 2015.**

The paper focuses on how to improve the antenna performance by the analysis of bandwidth enhancement. This paper is about distinct techniques that are available to feed microstrip patch antennas. These techniques can be contacting and non-contacting techniques. In the contacting technique, the RF power is fed directly to the radiating patch with the aid of a connecting element like a microstrip line. In the non-contacting technique, power is transferred between the microstrip line and the radiating patch via electromagnetic coupling. There are many feed techniques but microstrip line, coaxial probe, aperture coupling and proximity coupling are normally utilized to feed antenna. It was observed through simulations that concluded that the maximum bandwidth can be achieved by aperture coupling and proximity coupling because they provide best impedance matching and radiation efficiency. However, contacting techniques like edge feeding provide high gain and simple fabrication, i.e. reduced complexity.

**5. R. Kiruthika and T. Shanmuganantham, "Comparison of different shapes in microstrip patch antenna for X-band applications", International Conference on Emerging Technological Trends (ICETT), pp. 1-6, 2016.**

In this paper, a microstrip patch antenna for radar applications is presented for a variety of shapes available for the patch. In this paper, conventional shapes like Rectangular, Triangular and Circular Microstrip patch antenna are designed and analyzed. The antenna is designed to resonate at X-band frequency. The substrate used by the antenna is the low cost FR4 (Flame Retardant) Epoxy. The Ansoft HFSS (High Frequency Structural Simulator) Version 12 software is used to analyze the results of different shapes of Microstrip patch antenna. The parameters like return loss, bandwidth, gain and directivity are compared and discussed in this paper. From observation made via simulations, it is concluded that the rectangular microstrip patch antenna has an optimal result in all the parameters compared to that of the triangular and circular. But the triangular microstrip patch antenna provides narrow bandwidth with low gain and directivity. Even though circular patch antenna does not show better result than rectangular patch, it provides better antenna parameter characteristics to that of triangular. Hence, the rectangular microstrip patch antenna is well suited for X-band (Doppler radar) application having a frequency range from 8 to 12 GHz.

**6. M. Meena and P. Kannan, "Analysis of Microstrip Patch Antenna for four different shapes and substrates", ICTACT Journal on Microelectronics, Volume 04, Issue 01, April 2018.**

This paper presents an analysis on which substrate material is appropriate for designing a microstrip patch antenna based on its application. In this paper microstrip patch antenna is designed for four different shapes: H-shape, E-shape, S-shape and U-shape, and substrates: FR-4, RO4003, GML1000 and RT/Duroid 5880. The substrate materials are taken according to the dielectric constant or relative permittivity values.

Antenna parameters such as gain, directivity, bandwidth and return loss are simulated for different shapes and substrates using Advanced Design System (ADS) software. The proposed methodology revealed better performance in RT/Duroid5880 substrate material in terms of bandwidth and gives better gain in FR-4 substrate material for any shape. From the analysis it was found that the substrate materials which have lower dielectric constant values have provide the better bandwidth and the substrate materials which have higher dielectric constant values have provide the better gain and directivity.

**7. S. Huang, L. Zhang, C. Wang, C. Liu, S. Wu and X. Wu, "An axial-ratio beam-width enhancement of antenna based on electromagnetic bandgap (EBG)", 2020 IEEE 4th Information Technology, Networking, Electronic and Automation Control Conference (ITNEC), pp. 206-209, 2020.**

This paper proposes a low profile axial-ratio beam-width enhancement of antenna based on electromagnetic band gap (EBG), in order to meet the requirements of the navigation system for antennas with wide 3-dB Axial Ratio beam-width. The effect of surface wave on the circular polarization of the antenna is suppressed by loading two layers of mushroom-like EBG unit cells around the patch. The loading of the EBG increases the 3- dB AR beam-width of the antenna by 78% in the x-z plane and the by 84% in the y-z plane compared with the antenna without EBG, the simulation and test results demonstrate the effectiveness of this method. Two layers of mushroom-type electromagnetic band gap structure are loaded around the antenna to enhance the axial-ratio beam-width of the antenna. Compared with the antenna without EBG, the 3dB Axial Ratio beam-width of the antenna has been greatly improved.

**8. S Venkateshwara ;K K oteshwara Rao. ;Sambasiva Rao Kumbha. ;Design and implementation of circular polarized patch antenna using EBG Structure. IEEE 2022 Microwaves, Antennas, and Propagation Conference (MAPCON).DOI: 10.1109/MAPCON56011.2022.10047294.**

This research focuses on enhancing the performance of antennas in modern wireless communication devices while maintaining a low profile. Electromagnetic Band Gap (EBG) surfaces, engineered to improve antenna efficiency, are explored. Specifically, the study investigates a square patch antenna over a 2D-Mushroom EBG ground plane, comparing it with a conventional ground plane.The designed circular polarized patch antenna operates at 2.4 GHz, catering to ISM band applications. Notably, it features a square patch with truncated edges, showcasing Right Hand Circular Polarization (RHCP). The paper delves into the band gap characteristics of mushroom EBG unit cells, strategically placed between the patch and ground plane. This placement aims to eliminate back lobe effects and enhance overall antenna performance.The engineered mushroom EBG is designed to resonate at 2.45 GHz, with a forbidden band gap of 450 MHz, an in-phase reflection band gap of 390 MHz, and a high impedance of 3475 Ω at resonance. The introduction of the EBG ground plane significantly enhances gain to 5.35 dB, maintaining an excellent axial ratio at resonance, and achieves an operating bandwidth of 7%. Remarkably, these improvements come with a reduction in patch dimensionsThe design undergoes simulation in HFSS and practical testing using a VNA after fabrication. The results show a high correlation between simulation and testing, affirming the effectiveness of the engineered EBG ground plane in enhancing antenna performance.

After reviewing, several common drawbacks were observed in the research. Such as

1. Limited Bandwidth
2. Poor Performance in Multi-path Environments
3. Impedance Matching Issues
4. Size and Weight Constraints

By addressing these specific drawbacks, our project will contribute to advancing the state-of-the-art in Circular Polarized Patch Antenna Array with EBG structures, making them more efficient, practical, and suitable for a broader range of applications

# Chapter 3

# MICROSTRIP PATCH ANTENNA

## 3.1 Introduction

The conventional microstrip antenna consists of a pair of parallel conducting layers separated by a dielectric medium, known as substrate. In the configuration shown in Figure 3.1, the upper conducting layer, which is the patch, is the source of radiation where electromagnetic energy fringes off the edges of the patch and into the substrate. The lower conducting layer acts as a perfectly reflecting ground plane. Physically, the patch is a thin conductor that is an appreciable fraction of a wavelength in extent. The substrate layer thickness is 0.01– 0.05 of free-space wavelength (ho). This is used primarily to provide proper spacing and mechanical support between the patch and its ground plane.

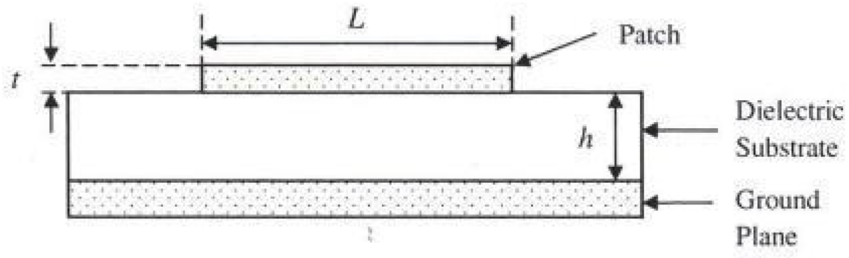


**Figure 3.1: Patch Antenna Cross Section**

## 3.2 Patch Antenna Fundamentals

### 3.2.1 Antenna Shapes

In order to simplify analysis and performance prediction, the patch is generally square, rectangular, circular, triangular, and elliptical or some other common shapes. For a rectangular patch, the length 'L' of the patch is usually 0.5 λo, where λo is the free-space wavelength. The patch is selected to be very thin such that t << λo, where 't' is the patch thickness. It was shown that rectangular microstrip antenna has an optimal result in all parameters compared to that of triangular and circular.

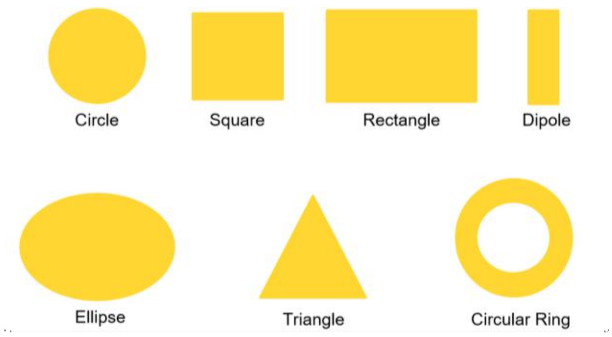


**Figure 3.2: Patch Antenna Shapes**

### 3.2.2 Advantages and Disadvantages of Patch Antenna

There are various antennas available, of which patch antennas are selectively preferable due to the various advantages it offers over other antenna types. Its low profile and compact nature makes it the optimal choice for antennas. However, it suffers from its own set of drawbacks. The Table 3.1 shows advantages and disadvantages.

**Table 3.1: Advantages and Disadvantages of Patch Antenna**

| S.No. | Advantages | Disadvantages |
|---|---|---|
| 1. | Light Weight & Low Volume | Narrow Bandwidth |
| 2. | Low Profile- Planar Configuration | Low efficiency |
| 3. | Low Fabrication Cost, High Production Volume | Low Gain |
| 4. | Both Linear & Circular Polarization are supported | Extraneous radiation from feeds |
| 5. | Easy integration with Microwave Circuits | Poor end fire radiator |
| 6. | Capable of dual and triple frequency operations | Low power handling capacity |
| 7. | Mechanically robust | Surface wave excitation |

### 3.2.3 Fringing Effect

A patch radiated from fringing fields around its edges. Electric field exists, but extends some distance away, not just directly between the conductive objects. The electric field distribution of rectangular patch excited in its fundamental mode is shown in Figure 3.3. The electric field is zero at the center of the patch, maximum (positive) at one side, and minimum (negative) on the opposite side. It should be mentioned that the minimum and maximum continuously change side according to the instantaneous phase of the applied signal. The electric field does not stop abruptly at the patch's periphery as in a cavity, rather the fields extend the outer periphery to some degree. These field extensions are known as fringing fields and cause the patch to radiate. A typical capacitor is composed of two conductive objects with a dielectric in between them. A voltage difference between the objects results in an electric field between them. In case of a parallel plate capacitor, the electric field exists not only between the planes, but extends some distance away, this is known as fringing field.

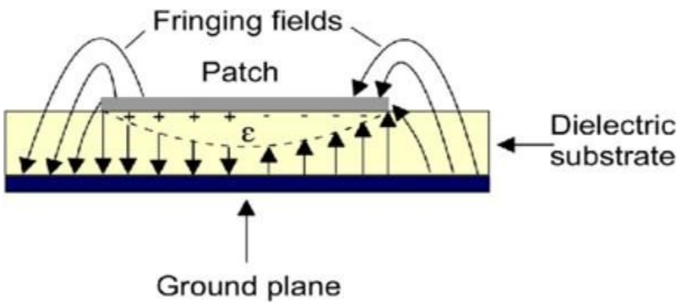


**Figure 3.3 : Fringing Fields in Patch**

## 3.3 Effect of Design Parameters

### 3.3.1 Dielectric Constant of Substrate

Dielectric constant of the substrate plays a significant role in patch antenna design. The substrate with high dielectric constant reduces antenna size but it also decreases the bandwidth (BW) and efficiency. A low value of dielectric constant for the substrate will increase the fringing field at patch periphery and thus radiated power. Thus, there is a trade off between size and performance of patch antenna.

### 3.3.2 Height of Substrate

The height of the substrate controls the bandwidth. Increasing the height increases the bandwidth because an antenna occupying more space in a spherical volume will have a wider bandwidth.The table indicates the substrate used.

| Substrate material | Dielectric constant ($\varepsilon_r$) |
|---|---|
| FR-4 | 4.4 |

**Table 3.2: Dielectric Substrate used**

A thicker substrate, besides being mechanically strong, will increase the radiated power, reduce conductor loss and the impedance bandwidth (BW) . However, it will also increase the weight, dielectric loss, surface wave loss and extraneous radiations from the feed.

### 3.3.3 Feed Point

Feed point refers to the location in patch where the input feed line is provided, in order to supply the input signal excitation to the radiating patch of the microstrip antenna. Selection of feed point plays a vital role in the performance of the antenna. It is vital for impedance matching between the generator and the radiating patch, in order to reduce reflections and increases the power radiated by the antenna.

## 3.4 Feeding Techniques

Feeding refers to the process of applying input excitation to the radiating patch of the microstrip antenna array in order for it to radiation energy. The four most repeatedly used feeding practices in antenna scheme are coaxial feed, microstrip line, aperture coupling and proximity coupling.

### 3.4.1 Microstrip Line Feed

The benefit of this style of feeding mechanism is patch as well as the feed is fabricated simultaneously on the same substrate, thus the antenna remains planar as shown in Figure 3.4. The disadvantage in edge-feed is that for impedance matching, a quarter wave transformer or similar structure is used thereby the length of feed increases to the size of patch and causing undesired radiations with increased cross-polar level.

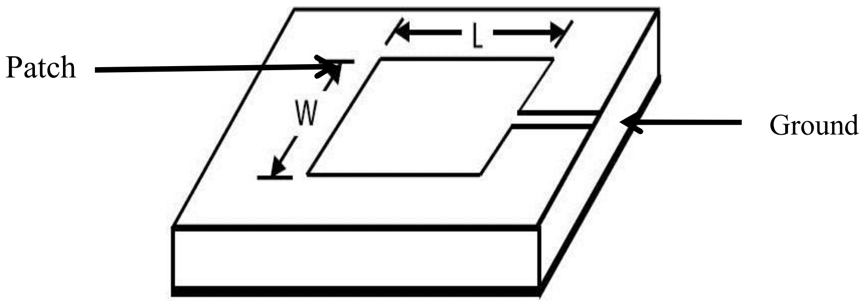


**Figure 3.4: Patch Antenna**

### 3.4.2 Quarter Wavelength Transmission Feed

The microstrip antenna can also be matched to a transmission line of characteristic impedance $Z_0$ by using a quarter-wavelength transmission line of characteristic impedance $Z_1$ as shown in Figure 3.5. The goal is to match the input impedance ($Z_{in}$) to the transmission line ($Z_0$). If the impedance of the antenna is $Z_A$, then the input impedance viewed from the beginning of the quarter-wavelength line

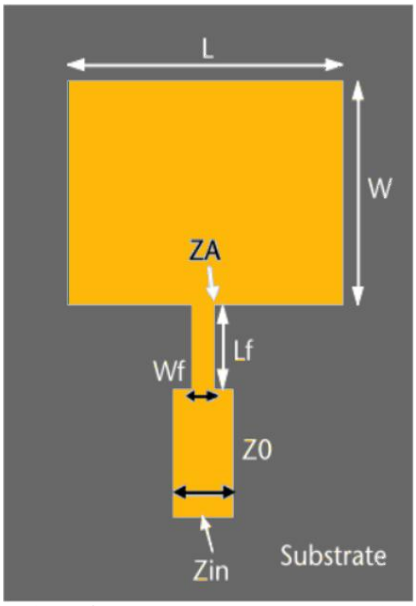


**Figure 3.5 : Quarter Wave Fed Patch Antenna**

This input impedance Zin can be altered by selection of the Z1, so that Zin=Z0 and the antenna is impedance matched. The parameter Z1 can be altered by changing the width of the quarter-wavelength strip. The wider the strip is, the lower the characteristic impedance (Z0) is for that section of line. Edge feed is used to feed the antenna. The edge impedance ZA of the patch antenna ..

### 3.4.3 Coaxial Probe

It is also one of the most commonly used methods to feed a microstripantenna. The inner conductor of the probe is in contact with the patch from the bottomside and the outer conductor to ground plane, the issue in this method is that a hole needs to be drilled carefully to connect the coaxial probe and mechanically this is not astrong and reliable connectionthe issue in this method is that a hole needs to be drilledcarefully to connect. The structure is shown in Figure 3.6

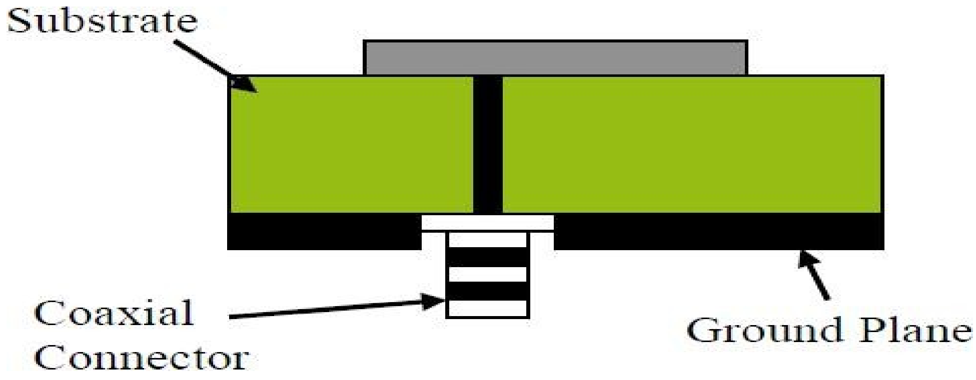


**Figure 3.6: Coaxial Feed**

### 3.4.4 Proximity Coupled

The substrate used here is of two layers as shown in Figure 3.7 where patch is on the upper layer and feed comes on the lower layer. The advantages of this type of feed are elimination of undesired radiations which are prominent in microstrip line feed case. This type of feed gives increased bandwidth due to increase in substrate thickness. The disadvantages of this technique are increased antenna size due to thicker substrate and alignment of two substrate layers and hence antenna fabrication becomes tedious.

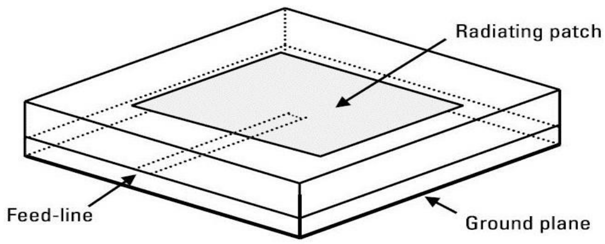


**Figure 3.7: Proximity Coupled Feed**

### 3.4.5 Aperture Coupled

The advantages of this feeding technique are wider bandwidth and shielding of patch antenna from unwanted radiations from feed structure thus maintaining polarization purity. In this coupling the field of feed is interacted with the patch through a slot in the ground plane as illustrated in Figure 3.8

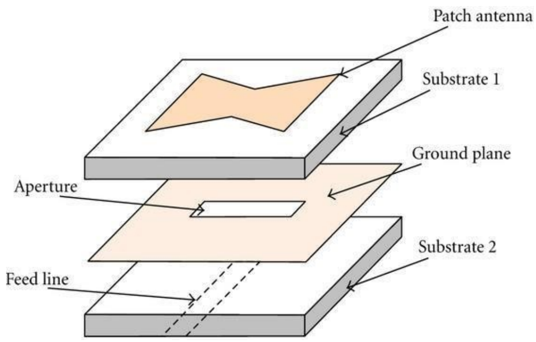


**Figure 3.8: Aperture Coupled Feed**

The bandwidth obtained is around 12%. Impedance matching can be accomplished by adjusting the measurement of feed line and size of slot used in the ground plane. The substrate used for the feed line should have high dielectric constant and should be thin. In depth analysis of this type of feeding technique can be found in references.The fabrication of antenna using this type of feeding mechanism is difficult.Some other techniques include Co-planar Wave Guide feeding. In this technique, both the microstrip patch and ground plane are modeled on the same side of the substrate material.

## 3.5 Circular Polarization

### 3.5.1 Introduction

Polarization of a radiating antenna refers to the orientation of net electric field vector with respect to a reference plane, which is generally considered to be the ground plane as shown in Figure 3.9. In a circularly-polarized antenna, the plane of polarization rotates in a corkscrew pattern making one complete revolution during each wavelength. A circularly- polarized wave radiates energy in the horizontal, vertical planes as well as every plane in between. The rotation can be either in clockwise or anti-clockwise direction. A circularly polarized microstrip antenna is categorized into two types by its feeding systems: one is a dual-feed CP antenna with an external polarizer and the other is a singly-fed one without a polarizer.

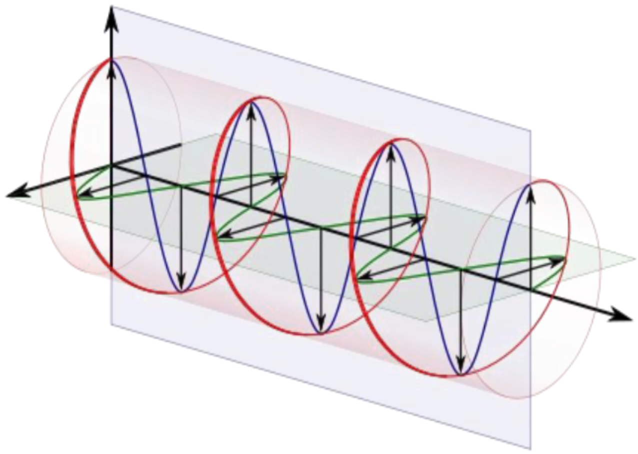

**Figure 3.9: Circular Polarization**

Nowadays, circular polarization is crucial in the antenna design industry as it eliminates the requirement of antenna orientation in the plane perpendicular to the propagation direction. It gives much more flexibility to the angle between transmitting and receiving antennas. Also, it enhances weather penetration and mobility.

### 3.5.2 Circular Polarization Technique

A single-fed CP antenna may be regarded as one of the simplest radiators for exciting circular polarization. According, the operational principle of this antenna is based on the fact that the generated mode can be separated into two orthogonal modes by the effect of perturbation segment such as a slot or other truncated segment.

Consequently, by setting the perturbation segment to the edge of the patch at 45° to the feed, the generated mode is separated into two orthogonal modes. The radiated fields excited by these two modes are perpendicular to each other, hence orthogonally is achieved which is the condition for circular polarization.

When the amount of perturbation segment is adjusted to the optimum value, modes 1 and 2 are exited in equal amplitude and 90 degrees out of phase at the center frequency. This allows the antenna to act as a circular polarized radiator in spite of single feeding.

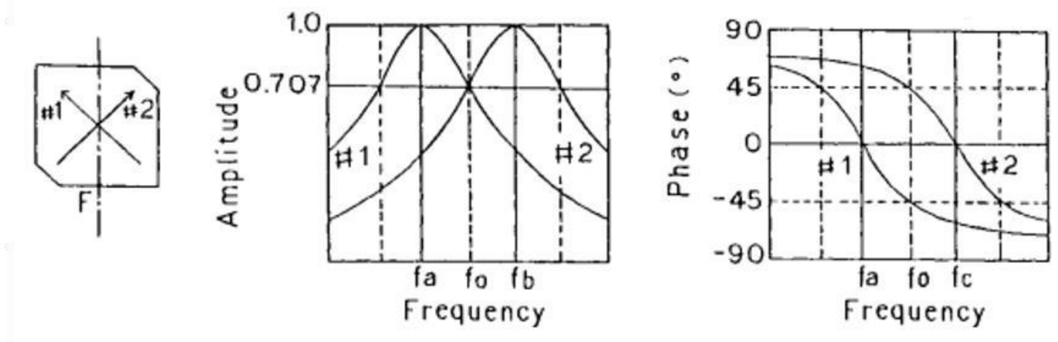


**Figure 3.10 : Circular Polarization Characteristics**

## 3.6 Drawbacks of Microstrip Antenna

The mechanisms of transmission and radiation in a microstrip can be understood by considering a point current source (Hertz dipole) located on top of the grounded dielectric substrate .This source radiates electromagnetic waves. Depending on the direction toward which waves are transmitted, they fall within three distinct categories as shown in Figure 3.11, each of which exhibits different behaviors.

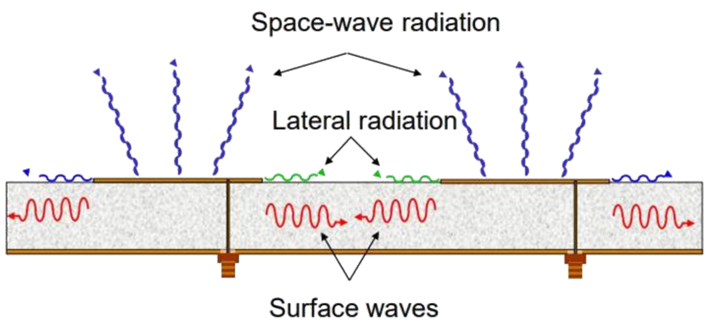


**Figure 3.11: Waves of Microstrip Antenna**

### 3.6.1 Surface Waves

The waves transmitted slightly downward, having elevation angles θ between π/2 and π-arcsin ($1/\sqrt{\varepsilon r}$), meet the ground plane, which reflects them, and then meet the dielectric-to-air boundary, which also reflects them (total reflection condition). Following this zig-zag path, they finally reach the boundaries of the microstrip structure where they are reflected back and diffracted by the edges giving rise to end-fire radiation. The magnitude of the field amplitudes builds up for some particular incidence angles that leads to the excitation of a discrete set of surface wave modes which are similar to the modes in metallic waveguide. On other way in the boundary, if there is any other antenna in proximity, the surface wave can become coupled into it. Surface wave are TM and TE modes of the substrate. These modes are characterized by waves attenuating in the transverse direction (normal to the antenna plane) and having a real propagation constant above the cut-off frequency. The fields remain mostly trapped within the dielectric, decaying exponentially above the interface. The vector α, pointing upward, indicates the direction of largest attenuation. The wave propagates horizontally along β, with little absorption in good quality dielectric as shown in Figure 3.12. The same guiding mechanism provides propagation within optical fibers.

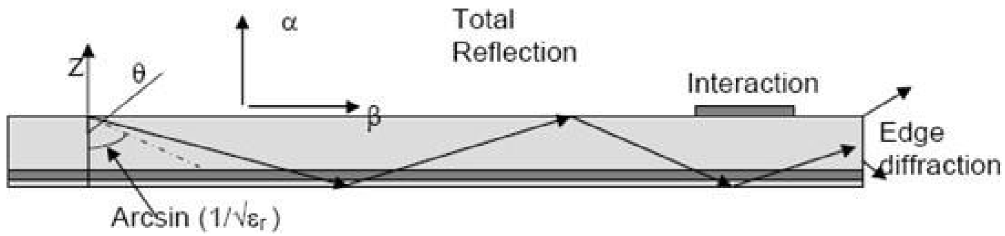


**Figure 3.12: Surface Waves**

### 3.6.2 Leaky Waves

Waves directed more sharply downward, with θ angles between π-arcsin ($1/\sqrt{\varepsilon_r}$) and π, are also reflected by the ground plane but only partially by the dielectric- to-air boundary. They progressively leak from the substrate into the air, hence their name leaky waves, and eventually contribute to radiation.

### 3.6.3 Guided Waves

When realizing printed circuits, one locally adds a metal layer on top of the substrate, which modifies the geometry, introducing an additional reflecting boundary.

Waves directed into the dielectric located under the upper conductor bounce back and forth on the metal boundaries, which form a parallel plate waveguide. The waves in the metallic guide can only exist for some Particular values of the angle of incidence, forming a discrete set of waveguide modes. The guided waves provide the normal operation of all transmission lines and circuits, in which the electromagnetic fields are mostly concentrated in the volume below the upper conductor.

## 3.7 Antenna Parameters

### 3.7.1 Return Loss

Return loss is the power of the reflected signal in a transmission line. It is defined as ratio of reflected power ($P_r$) and transmitted power ($P_t$) and expressed in decibels. For maximum power transfer, the return loss should be to as small as possiblewhich means that the ratio $P_r/P_t$ should be as small as possible. Return loss is a measure of reflection from an antenna. 0dB means that all the power is reflected, hence matching is not good. -10dB indicates that 10% of incident power is reflected meaning 90% of power is accepted by the antenna. So, having -10dB as a bandwidth reference is an assumption that 10% of the energy is lost. Return loss is determined in dB.

$$R_L = -20\log|\Gamma| dB \qquad \text{Eq.3.1}$$

Where, $|\Gamma|$ is reflection coefficient:

$$|\Gamma| = \frac{V_{0-}}{V_{0+}} = \frac{Z_L - Z_0}{Z_L - Z_0} \qquad \text{Eq.3.2}$$

$V_0^-$ = Reflected Voltage

$V_0^+$ = Transmitted Voltage

$Z_l$ = Load Impedance

$Z_0$ = Characteristics Impedance

### 3.7.2 VSWR

VSWR is the ratio of the maximum to minimum voltage on the antenna feeding line. The Voltage Standing Wave Ratio (VSWR) is an indication of how good the impedance match is. For perfect impedance matched antenna the VSWR is 1. VSWR causes return loss or loss of forward energy through a system, when an antenna and feed line do not have matching impedances, some of the electrical energy cannot be transferred from the feed line to the antenna. Energy not transferred to the antenna is reflected back towards the transmitter. It is the interaction of these reflected waves with forward waves which cause standing wave patterns. Where |r| is reflection coefficient and Vmax and Vmin are maximum and minimum voltages respectively. These voltages are obtained from the standing waves formed due to impedance mismatch.

The equation is defined as:

$$VSWR = \frac{V_{max}}{V_{min}} = \frac{1+|r|}{1-|r|} \qquad \text{Eq.3.3}$$

### 3.7.3 Bandwidth

The bandwidth of the antenna said to be those range of frequencies within which the performance of the antenna conforms to a specified standard. Bandwidth is the function of resonant frequency. The bandwidth can be calculated from return loss plot, called Impedance Bandwidth. The bandwidth at -10dB at lower and upper frequency from return loss plot is considered. The bandwidth of narrow band and broadband antennas are defined as where $F_h$ is the upper frequency,$F_l$ is the lower frequency

$$BW = Fh - Fl \qquad \text{Eq.3.4}$$

### 3.7.4 Radiation Pattern

A radiation pattern defines the variation of the power radiated by an antenna as a function of the direction away from the antenna. This power variation as a function of the arrival angle is observed in the antenna's far field. The radiation pattern is mathematical function or graphical depiction of the relative field strength transmitted from or received by the antenna. If the total power radiated by the isotropic antenna is P, then the power is spread over a sphere of radius.

**Main Lobe**

This is the radiation lobe containing the direction of maximum radiation.

**Side Lobe**

These are the minor lobes adjacent to the main lobe and are separated by various nulls. Side lobes are generally the largest among the minor lobes.

**Back Lobe**

This is the minor lobe diametrically opposite the main lobe. Represents radiation in undesirable direction. It increases noise levels of a system.

### 3.7.5 Gain

Gain is the ratio of intensity, in a given direction, to the radiation intensity that would be obtained if the power accepted by the antenna were radiated isotropically.

$$\text{Gain} = 4\pi \frac{\text{Radiation Intensity of Test Antenna}}{\text{Isotropic Reference Antenna with same Input Power}} = 4\pi \frac{U(\theta,\emptyset)}{P_{in}} \qquad \text{Eq.3.5}$$

Isotropic antenna is considered to have a gain of unity. The gain function can be described as:

$$G(\theta,\emptyset)=\frac{P(\theta,\emptyset)}{\frac{Wt}{4\pi}} \qquad \text{Eq.3.6}$$

Where $P(\theta,\emptyset)$ is the power radiated per unit solid angle in the direction $(\theta,\emptyset)$ and Wt is the total radiated power.

### 3.7.6 Directivity

The directivity of an antenna is defined as the measure of concentrated power radiated in a particular direction. It may be considered as the capability of an antenna to direct radiated power in a given direction. It can may be considered as the capability of an antenna to direct radiated power in a given direction also be noted as the ratio of the radiation intensity in a given direction to average radiation intensity.

$$D=\frac{U}{U_o} = \frac{4\pi U}{P_{rad}} \qquad \text{Eq.3.7}$$

If a three dimensional antenna pattern is measured, the ratio of normalized power density at the peak of the main beam to the average power density is called the directivity. If a three dimensional antenna pattern is measured, the ratio of normalized power density at the peak of the main beam to the average power density is called the directivity. The relation between directivity and gain can be given .

$$G=\eta D \qquad \text{Eq.3.8}$$

Where $\eta$ is the antenna efficiency.

### 3.7.7 Radiation Efficiency

The radiation efficiency of the patch antenna is affected not only by conductor and dielectric losses, but also by surface-wave excitation, since the dominant $TM_{01}$ mode of the grounded substrate will be excited by the patch.

Where, $P_r$ radiated power and $P_i$ input power

$P_i = P_r + P_c + P_d + P_{sw}$

$P_c$ = Power Loss due to Conducting Patch

$P_d$ = Power Loss due to Dielectric Substrate

$P_{sw}$ = Power Loss due to Surface Wave Excitation

**3.7.8 Polarization**

Polarization is a very important parameter when choosing and installing an antenna. The polarization of an antenna refers to the orientation of the electric field (E- plane) of the radio wave with respect to the Earth's surface and is determined by the physical structure of the antenna and by its orientation.

A linear polarized antenna radiates wholly in one plane containing the direction of Propagation i.e. the electric field and the magnetic field are perpendicular to each other and to the direction the plane wave is propagating.

In a circular polarized antenna, the plane of polarization rotates in a circle making one complete revolution during one period of the wave. Circular polarization can be obtained if two orthogonal modes are excited with a 90 degrees time-phase difference between them. This can be accomplished by adjusting the Physical dimensions of the patch and using either single, or two, or more feeds. Also, the electric-field vector at a given point in space traces a circle as a function of time. The conditions to accomplish this are if the electric field vector possesses all the following:

1. The field must have two orthogonal linear components.
2. The two components must have the same magnitude.
3. The two components should have a time-phase difference of odd multiples of 90 degrees.
4. Elliptical Polarization is achieved when the E-field has two perpendicular components that are out of phase by 90 degrees but are not equal in magnitude. Hence, the net field rotates as in case of circular polarization but the magnitude is not constantThe field must have two orthogonal linear components.
5. The two components must have the same magnitude.
6. The two components should have a time-phase difference of odd multiples of 90 degrees.

7. The field must have two orthogonal linear components.
8. The two components must have the same magnitude.
9. The two components should have a time-phase difference of odd multiples of 90 degrees.

10. The field must have two orthogonal linear components.
11. The two components must have the same magnitude.
12. The two components should have a time-phase difference of odd multiples of 90 degrees.

Elliptical Polarization is achieved when the E-field has two perpendicular components that are out of phase by 90 degrees but are not equal in magnitude. Hence, the net field rotates as in case of circular polarization but the magnitude is not constant.

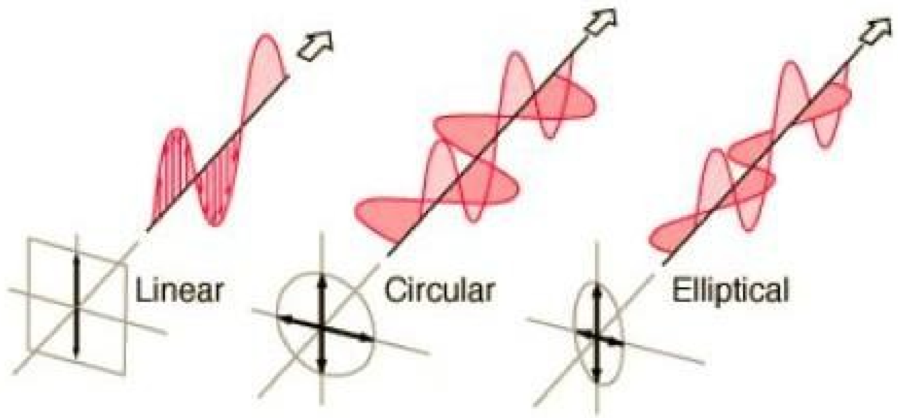


**Figure 3.13 : Types of Polarization**

**Axial Ratio**

The axial ratio is the ratio of orthogonal components of electric field vectors. It is also defined as the ratio of horizontal field component of electric field to that of vertical field component. A circularly polarized field is made up of two orthogonal E- field components of equal amplitudes (and 90 degrees out of phase). Because the components are equal magnitude, the Axial Ratio is 1 (or 0 dB). The Axial Ratio (AR) can we written as:

$$AR = \frac{\text{Major Axis}}{\text{Minor Axis}} \quad 1 \leq AR \leq \infty \qquad \text{Eq.3.10}$$

**Axial Ratio Bandwidth**

This bandwidth is related to the polarization bandwidth and this number expresses the quality of the circular polarization of an antenna. It is measured at 3dB level in the axial ratio plot. It can be defined as the frequency range at which axial ratio is less than 3dB acceptable level as seen in Figure 3.14.

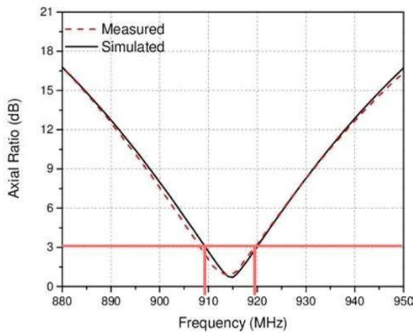


**Figure 3.14 : Axial Ratio Plot with Bandwidth**

### 3.7.9 Smith Chart

The Smith Chart is a fantastic tool for visualizing the impedance of a transmission line and antenna system as a function of frequency which is shown in Figure 3.15.

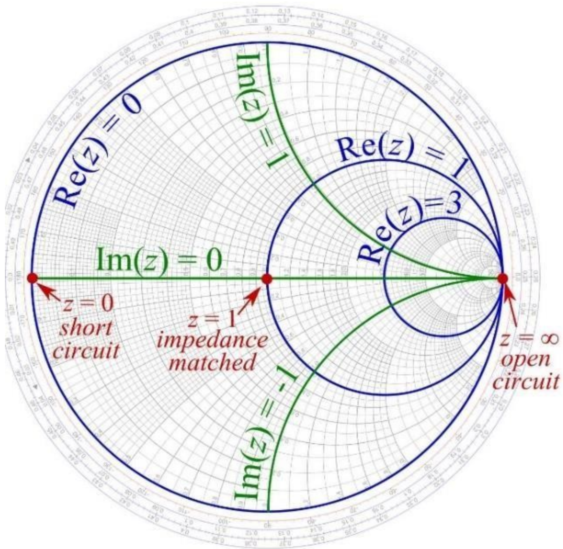


**Figure 3.15 : Smith Chart**

Smith Charts are usually normalized to 50 ohms The Smith chart is plotted on the complex reflection coefficient plane in two dimensions and is scaled in normalized impedance.

# Chapter 4

# HIGH IMPEDANCE SURFACE - EBG

High impedance surface like Electromagnetic Band Gap (EBG) materials are a class of artificial materials that exhibit unique properties not found in naturally occurring substances. Similar to metamaterials, EBG materials are engineered to manipulate electromagnetic waves in specific ways. These materials are designed to create a band gap, a range of frequencies in which electromagnetic waves cannot propagate through the material. Thischaracteristic allows EBG materials to control and manipulate electromagnetic wave propagation, making them useful in various applications such as antennas, microwave circuits, and communication systems. The ability to engineer these properties opens up new possibilities in the fields of photonics and radio frequency engineering, enabling advancements in technology that leverage the tailored control of electromagnetic waves.

## 4.1 Introduction

Electromagnetic Band Gap (EBG) is also referred to as Photonic Band Gap (PBG) high impedance surface. This structure is compact which has good potential to build low profile and high efficiency antenna surface. The main advantage of EBG structure is their ability to suppress the surface wave current. The generation of surface waves decreases the antenna efficiency and degrades the antenna pattern. Furthermore, it increases the mutual coupling of the antenna array which causes the blind angle of a scanning array. The feature of surface-wave suppression helps to improve antenna's performance such as increasing the antenna gain and less power wasted when reducing backward direction. This project work focuses on Mushroom-type two dimensional EBG structure. In this project, the antenna substrate is periodically loaded so that the surface wave dispersion diagram presents a forbidden frequency range about the antenna operating frequency. Because the surface waves cannot propagate along the substrate, an increase amount of radiating power couples to the space waves as shown in Figure 4.1. Also, other surface wave coupling effects like mutual coupling between array elements and interference with on-board systems are eliminated .

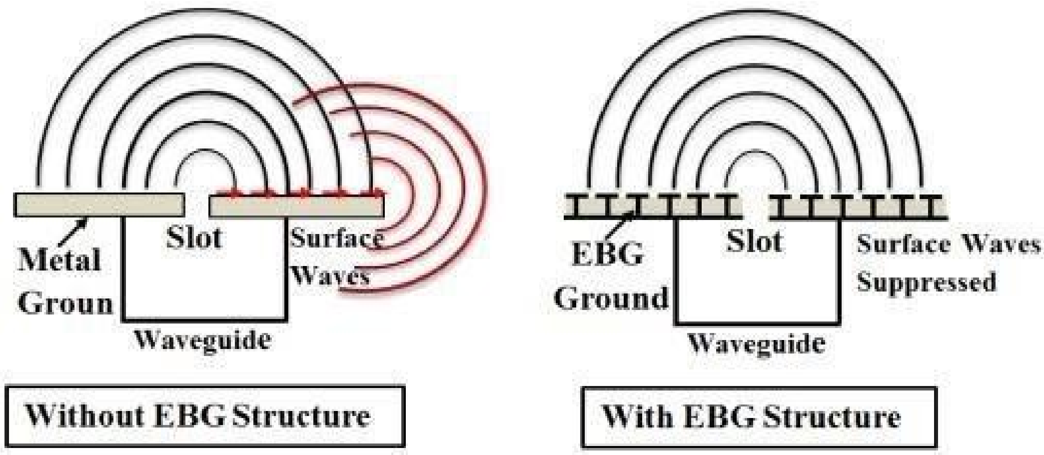


**Figure 4.1: Surface Waves Diffraction**

The reflection-phase of the EBG structure is in the range of -90 to +90 degrees, which produces addition of the incident and reflected signals in a quasi-constructive fashion. At 0 degree reflection, the EBG structure resembles the behavior of artificial magnetic conductors .

EBG configurations are differentiated into one, two, and three-dimensional formats. Due to its ease of fabrication, the two-dimensional EBG model is the focus of this discussion. This model's functionality can be understood through an effective medium approach, where it's represented by an equivalent network of inductors (L) and capacitors (C). Each EBG unit cell typically consists of a metallic patch at top a substrate, a bottom ground plane, and a metallic via that connects the patch directly to the ground plane. The space separating the patches introduces capacitive properties, whereas the via induces inductive properties.

### 4.1.1 Analysis of Mushroom EBG structure

The mushroom-like EBG structure is a 2-D EBG surface that consists of a metallic patch and a via that connects the patch to the ground as shown in Figure 4.2. The operating mechanism of the structure can be explained using an effective medium model with equivalent lumped LC elements. A unit cell EBG structure is composed of a metallic patch, a substrate, a ground plane, and a metallic via connecting the patch to the ground plane. The gap between adjacent patches has the effect of capacitance while the current flowing through the metallic via has an inductive effect. At its resonant frequency, the impedance is very high, and the structure impedes any surface waves, resulting in a frequency bandgap.

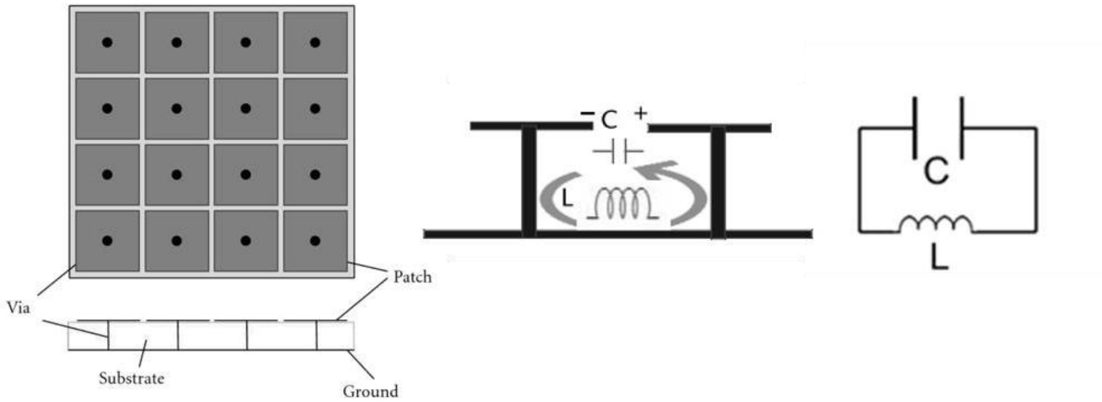


**Figure 4.2: Mushroom-like EBG structure and equivalent model**

# Chapter 5

# PROPOSED ANTENNA DESIGN

## 5.1 Introduction

A microstrip patch antenna array consists of a substrate dielectric material sandwiched between a patch and a ground plane made of conducting material like copper or gold, with either contacting or non-contacting feeding for patch excitation.

## 5.2 Proposed Design

A rectangular patch with Quarter Wavelength feeding is chosen for better impedance matching with the patch to attain good return loss, and to provide input excitation to the patch. Circular Polarization in antenna is achieved by truncating two opposite edges of the patch which generates two orthogonal degenerative modes of excitation necessary for generating two electric field components orthogonal to one another.The extent of each truncation is carefully adjusted to ensure that the phase of the secondary resonance is orthogonal to the primary, a requirement for achieving circular polarization effectively. Geometric alterations from the edge truncation result in non-identical diagonal lengths of the patch, which are instrumental in producing the dual resonances observed.

The 2*2 antenna array is integrated with a periodic Electromagnetic Band Gap (EBG) structure serving as the ground plane, which significantly enhances the antenna's performance. The design optimizes the EBG's properties to operate effectively within its band gap, minimizing the impact of surface waves by attenuating those that encounter it. Ideally, the inclusion of EBG structures is aimed at reducing undesirable effects such as mutual coupling and radiation from the back lobe, which are common challenges in antenna design.This periodic structure forms an EBG ground plane beneath radiating patch, since the mushroom patch is shorted to ground with stubs or vias. It behaved like an Artificial Magnetic Conductor. Their purpose is to minimize the surface waves by exhibiting Band Stop behavior, acting as a Perfect Magnetic conductor. The proposed design is shown in Figure

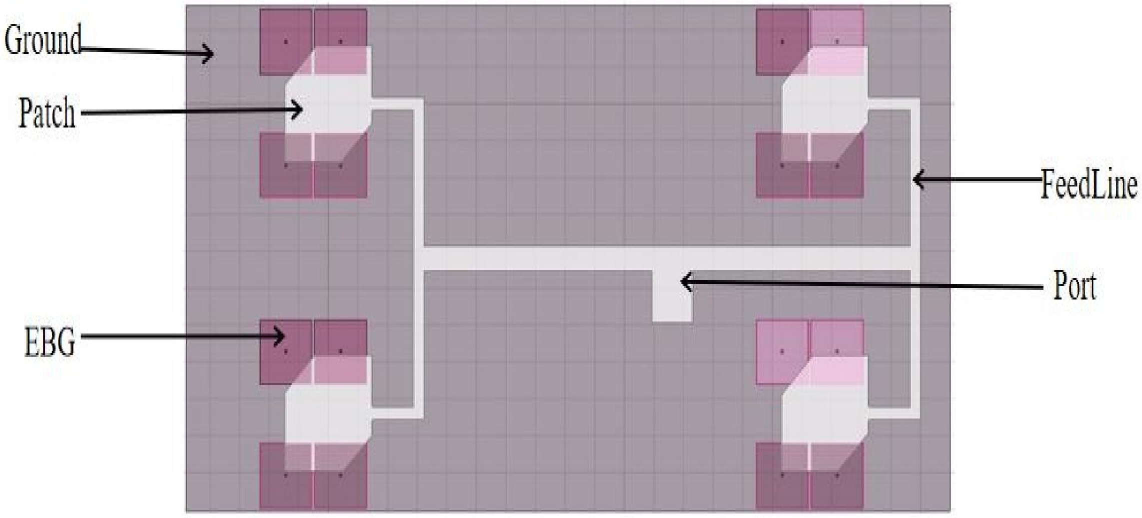


**Figure 5.1: Proposed Microstrip Patch Antenna Array with EBG (Top View)**

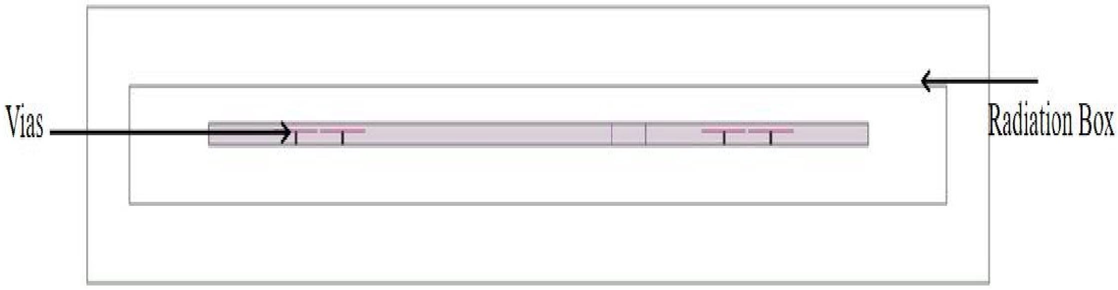


**Figure 5.2: Proposed Microstrip Patch Antenna Array with EBG (Side View)**

To further increase the efficiency of the antenna and improve its performance characteristics, two-dimensional periodic structures are introduced in the antenna design underneath the patch and within the substrate, called as Mushroom type Electromagnetic Band Gap structure. This periodic structure forms an EBG ground plane beneath radiating patch, since the mushroom patch is shorted to ground with stubs or vias. It behaved like an Artificial Magnetic Conductor. Their purpose is to minimize the surface waves by exhibiting Band Stop behavior, acting as a Perfect Magnetic Conductor (PMC), by rejecting or attenuating all surface wave components at the frequency range of operation of patch antenna. They also provide quasi in phase reflections to the plane waves incident from the patch onto the EBG plane

## 5.3 Microstrip Patch Antenna Array Design

Rectangular microstrip patch antennas, in their conventional form, are narrow-band structures. Their impedance bandwidth is typically 1–2%. This can be attributed to two factors: the resonant style of antenna (which makes the antenna radiate efficiently only over a narrow band of frequencies) and the thin thickness of antenna, typically less than 0.05λ0. This feature of conventional microstrip patch antennas makes them unsuitable in many applications where quite wide bandwidth is required. So, many researches have been done during the past few decades to overcome this limitation and several procedures have been proposed are need to be determined, the necessary equations for designing the rectangular patch are given below:

Next a 2x2 antenna array comprises four antenna elements arranged in a square grid, enhancing both performance and directional control compared to a single antenna. This configuration is widely used due to its balance between complexity and performance enhancement, making it ideal for numerous applications in communications and radar systems. It provides advantages such as enhanced gain and directivity, beamforming capabilities, simplified design and implementation. Overall, a 2x2 antenna array offers a practical solution for enhancing antenna performance with manageable complexity. It provides significant improvements ingain, directivity, and beam steering capabilities, making it an excellent choice.

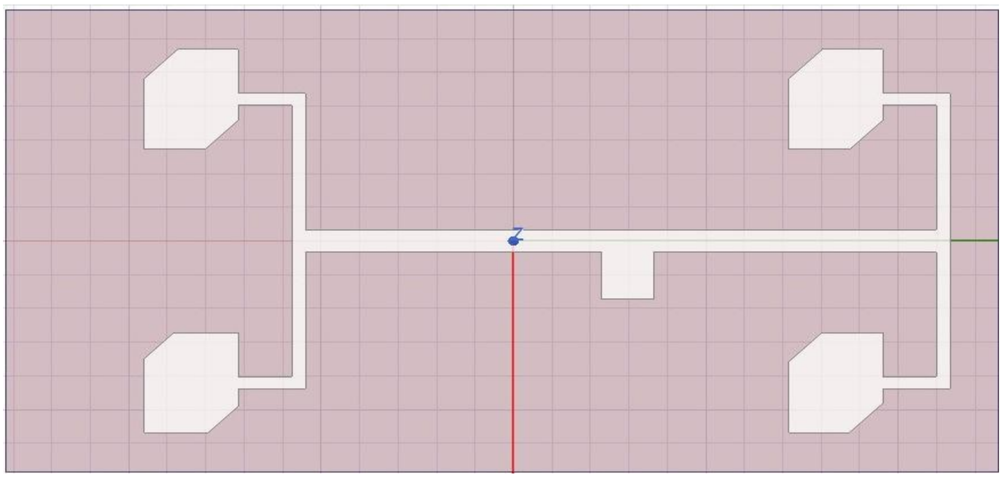

**Figure 5.3: Patch Antenna Array Design.**

## Chapter 6
# SOFTWARE IMPLEMENTATION

## 6.1 Introduction

Software simulation model is a new technology to yield high accuracy analysis and design of complicated microwave and RF printed circuits. Today there are a number of Electromagnetic modeling tools available in market such as HFSS, CST, etc. In this project, we have opted for HFSS (High Frequency Structure Simulator) tool to design and verify the characteristics of desired patch antenna. This chapter provides in depth details and familiarizes the user interface and options of HFSS.

## 6.2 Simulation Software- HFSS

HFSS stands for High Frequency Structure Simulator, whose logo is in Figure 6.1. It is an electromagnetic simulation software used for designing and simulating various electronics products. It is essential for designing and simulating high-frequency electronic products such as antennas, antenna arrays, RF or microwave components, high-speed interconnects, filters, connectors, IC packages and printed circuit boards. Engineers worldwide use Ansys HFSS to design high-freq uency, high-speed electronics found in communications systems, radar systems, advanced driver assistance systems (ADAS),satellites, internet-of-things (IoT) products and other high-speed RF.


**Figure 6.1: Ansys HFSS Logo**

It integrates simulation, visualization, solid modeling, and automation in an easy to learn environment where solutions to all types of 3D EM problems are quickly and accurate obtained. Ansys HFSS employs the Finite Element Method (FEM), adaptive meshing, and brilliant graphics to give unparalleled performance and insight to all types of 3D EM problems. Ansys HFSS can be used to calculate parameters such as S-Parameters, Resonant Frequency, and Fields. Ansys HFSS has evolved over a period.

## 6.3 Flowchart

The flow chart in Figure 6.2 depicts the sequence of steps to be followed for design, verification, fabrication and testing of the target microstrip patch antenna.

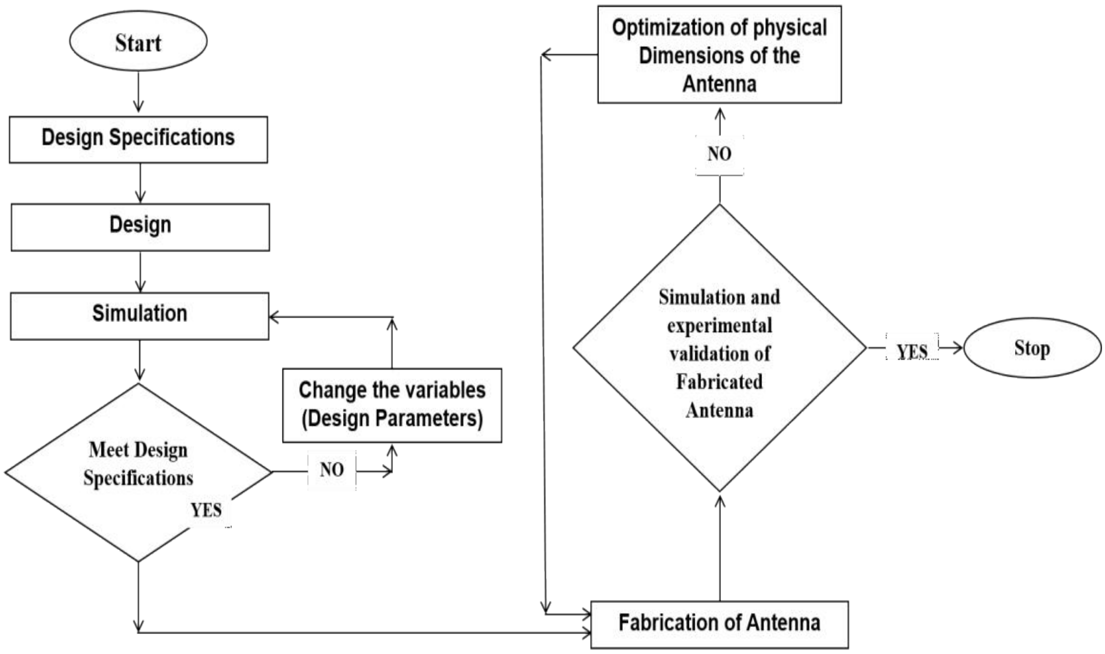


**Figure 6.2: Flowchart**

Starting with design specifications step, it involves selection of various aspects of design technologies such as type of antenna, type of feeding technique, polarization, antenna dimensions, substrate type and frequency range of operation. The next step is to design the antenna in a simulation tool, which in our case is HFSS (High Frequency Structure Simulator). The antenna is modelled to the desired specification, which depicts the design phase. This step is followed by simulation, which involves functional verification of antenna characteristics. Some vital antenna parameters are Return Loss (S11), Gain, Radiation Pattern, Antenna Efficiency, Axial Ratio, Impedance (Smith Chart), Bandwidth and Directivity. The next step is to check whether the simulation results are meeting the desired specifications. If not, the design parameter variables are altered and simulated again. If the design specifications are met, then the antenna design moves to fabrication phase, which involves manufacturing of the antenna. The fabricated antenna is then simulated with hardware and experimental testing and validation is performed in anechoic chamber with an antenna test setup and Vector Network Analyzer (VNA). If the result matches the desired specification with acceptable level of deviation, it is ready for its application. If not, the physical dimensions of antenna and other parameters are varied for correction and fabricated again. This process is repeated till the antenna is ready for its application.

## 6.4 Simulation Steps

Every HFSS simulation will involve the steps shown in Figure 6.3. While it is not necessary to follow these steps in exact order, it is good modeling practice to follow them in a consistent model-to-model manner. There are six main steps to creating and solving a proper HFSS simulation.


**Figure 6.3: Simulation Steps**

### 1. Create Model or Geometry

The initial task in creating an HFSS model consists of the creation of the physical model that a user wishes to analyze. This model creation can be done within HFSS using the 3D modeler. The 3D modeler is fully parametric and will allow a user to create a structure that is variable with regard to geometric dimensions and material properties. A parametric structure, therefore, is very useful when final dimensions are not known or design is to be tuned. The design for a quarter wave fed patch antenna is shown in Figure 4.4.

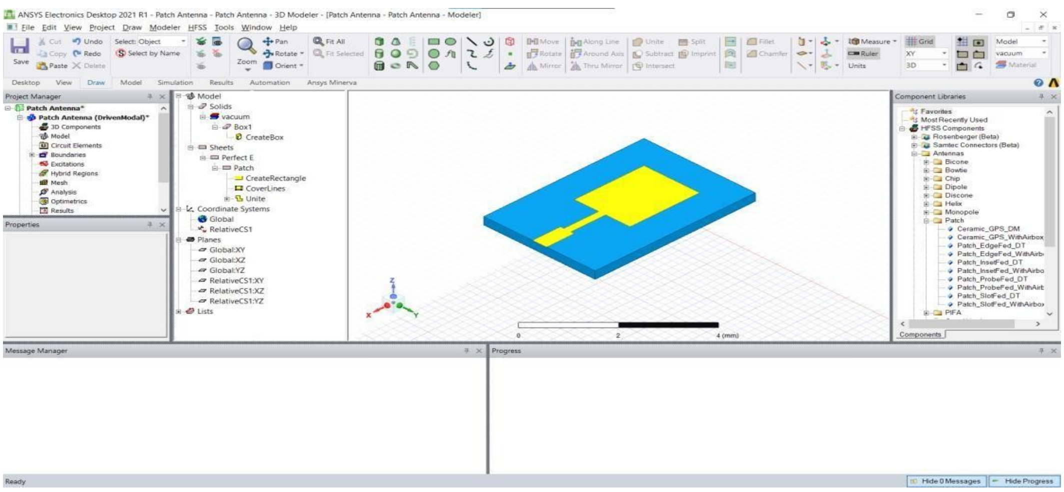

**Figure 6.4: User Interface**

The steps to create the above patch antenna structure is depicted in the Figure 6.5. In general, similar steps are followed for the modelling of any type of patch antenna. The yellow layer depicts the copper patch whereas the blue layer represents the substrate. The ground plane, substrate and the patch are stacked on top of each other. Material assignment should be performed properly in order to get accurate analysis results and antenna parameters.

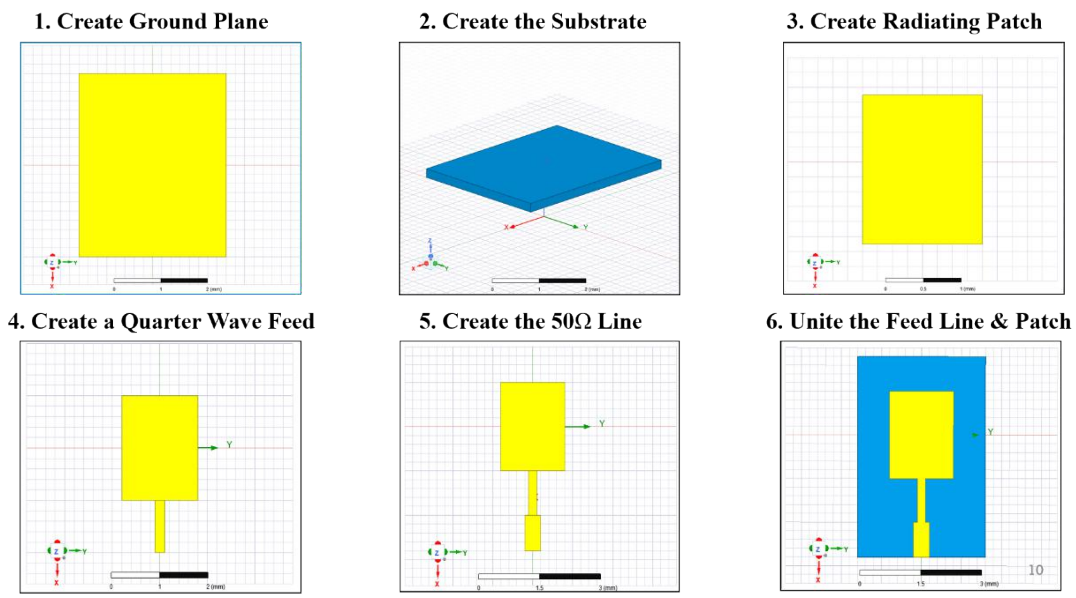


**Figure 6.5: Design Steps**

## 2. Assign Boundaries

The next crucial step is the assignment of boundaries. Boundaries are applied to specifically created 2D (sheet) objects or specific surfaces of 3D objects. Boundaries have a direct impact on the solutions that HFSS provides.

## 3. Assign Excitations

After the boundaries have been assigned as show in Figure 6.7, the excitations or ports should be applied in order to supply power to the antenna. The most commonly used excitation sources are Lumped Port and Wave Port. The Lumped Port deals with input in the form of voltage and current variations, whereas the Wave Port provides

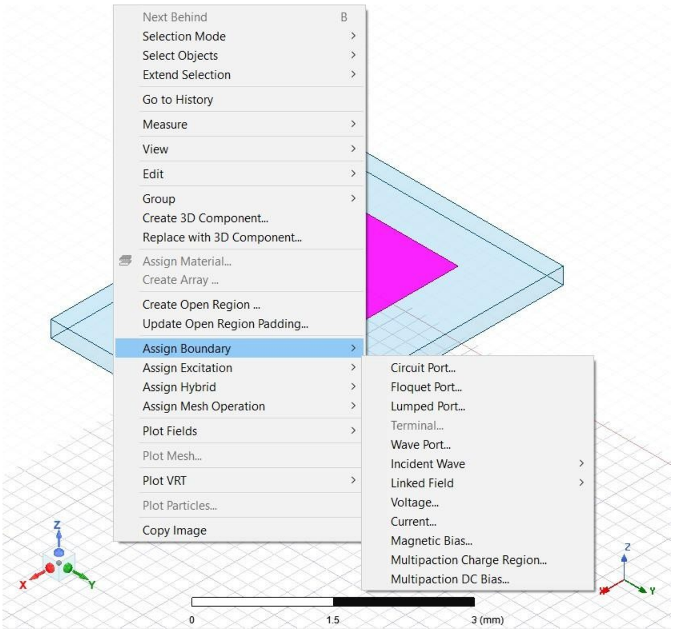


**Figure 6.7: Assign Excitation**

**4. Set up the Solution**

Solution setup involves setting the solution frequency for our design as shown in Figure 6.8. Frequency sweep defines the range of frequencies over which the antenna is simulated and necessary step size is added to achieve suitable results. A solution frequency of 2.4 GHz is set as shown in Figure.

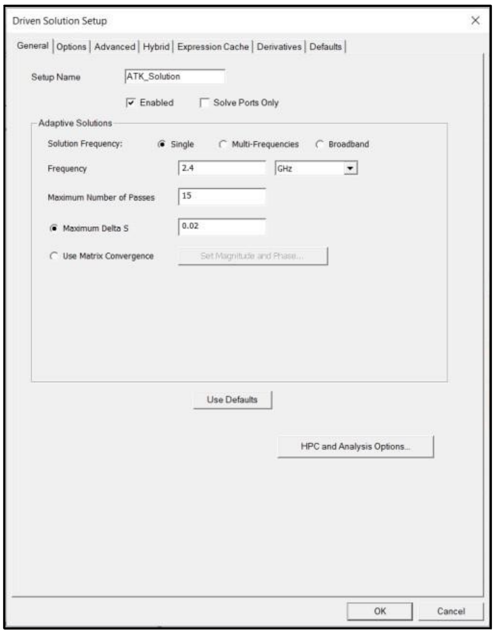


**Figure 6.8: Solution Setup**

The next step is to set the sweep. There are three types of sweep available in HFSS:

**Discrete Sweep**

The discrete sweep is the most accurate since it resolves the problem at each frequency point in your frequency bandwidth. However, it is the slowest of them all.

**Fast Sweep**

The fast sweep requires less simulation time than the discrete sweep but at the expense of a limited frequency bandwidth range and reduced accuracy.

**Interpolation Sweep**

The interpolating sweep is the fastest, but the least accurate sweep. In addition, the field data cannot be saved for the frequency range.

### 5. Solve

Solving of an HFSS projects includes validation of all the assignments, excitations, radiation boundaries and frequency setup. If any missing data it is displayed else validation check is completed as shown in below Figure 6.9. The model is ready to be analyzed which takes time based on the complexity of design.

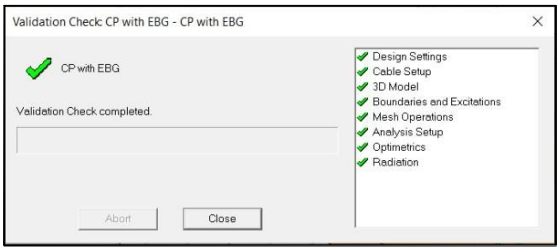


**Figure 6.9: Validation**

### 6. Post-Process the Results

After successful validation of the HFSS file and its execution, we need to post -process the results.If multiple solution setups are added, it is possible to generate results to them individually. The dialog box shown below represents the result generation interface for selecting the parameter.

## 6.5 Simulated Antenna Designs

In this project, an antenna designs are proposed at 2.4 GHz resonant frequency applications.Antenna basically incorporates truncated edges with EBG ground plane, and it consists of an array with is basically 2*2 size so this is why the antenna is called as Circular Polarized Patch Antenna Array with EBG structure. This antenna design serves the objectives of the project work.

### 6.5.1 Circular Polarized Patch Antenna Array with EBG structures

The Figure 6.10-6.11 shows the antenna design resonant at 2.4 GHz ISM band. This patch antenna design incorporates the periodic EBG ground plane. This antenna design exhibits circular polarization due to edge truncation and it is structured as 2*2 array. The performance parameters of this antenna design are analyzed and tabulated in the result section.

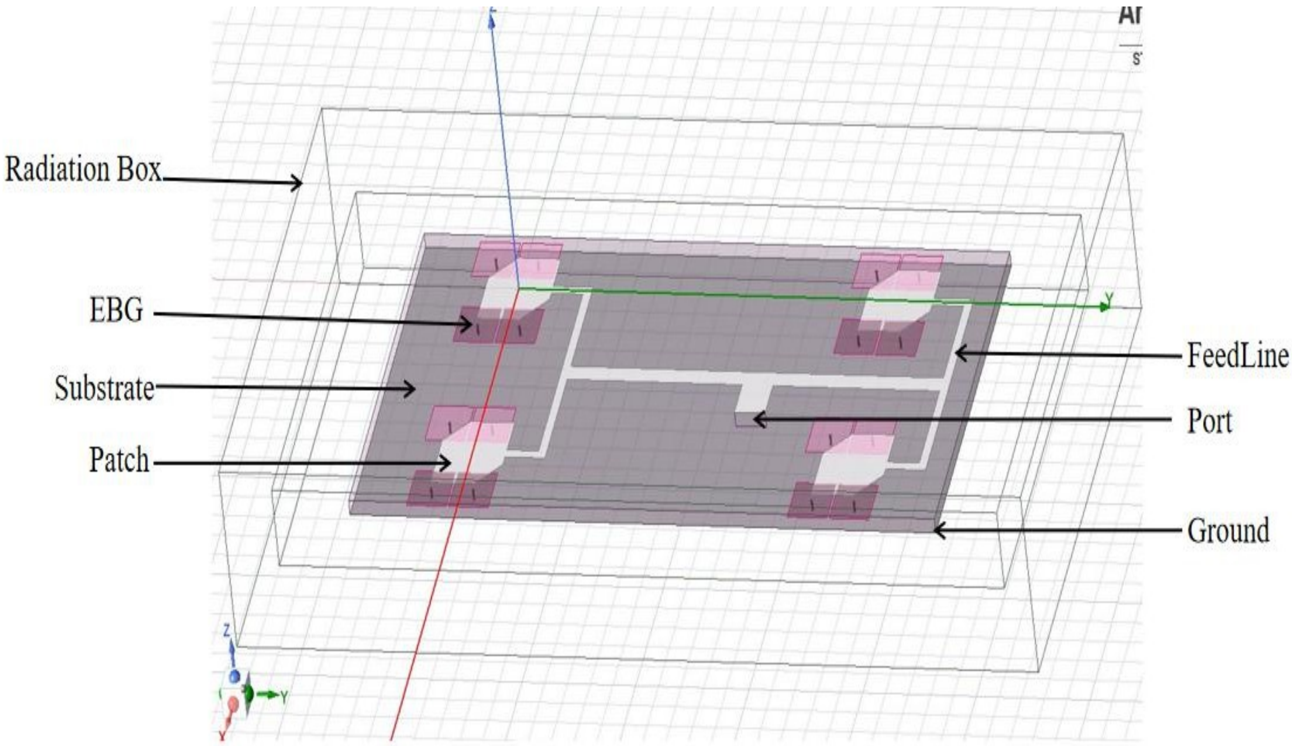


**Figure 6.10: Simulated Circular Polarized Patch Antenna Array with EBG**

### 6.5.2 EBG Unit Cell

The Figure 6.13 depicts the EBG Unit Cell that is designed to resonate at around 2.4 GHz. Instead of simulating all the periodic EBG structure forming the ground plane at once, we analyze a single EBG unit cell. This is done in order to reduce the complexity of simulation. Since the EBG ground plane is periodic, the same characteristic of unit cell is exhibited throughout the structure. Sufficient symmetry measures are taken in order to emulate the periodic presence of EBG cells at the neighborhood of each unit cell.

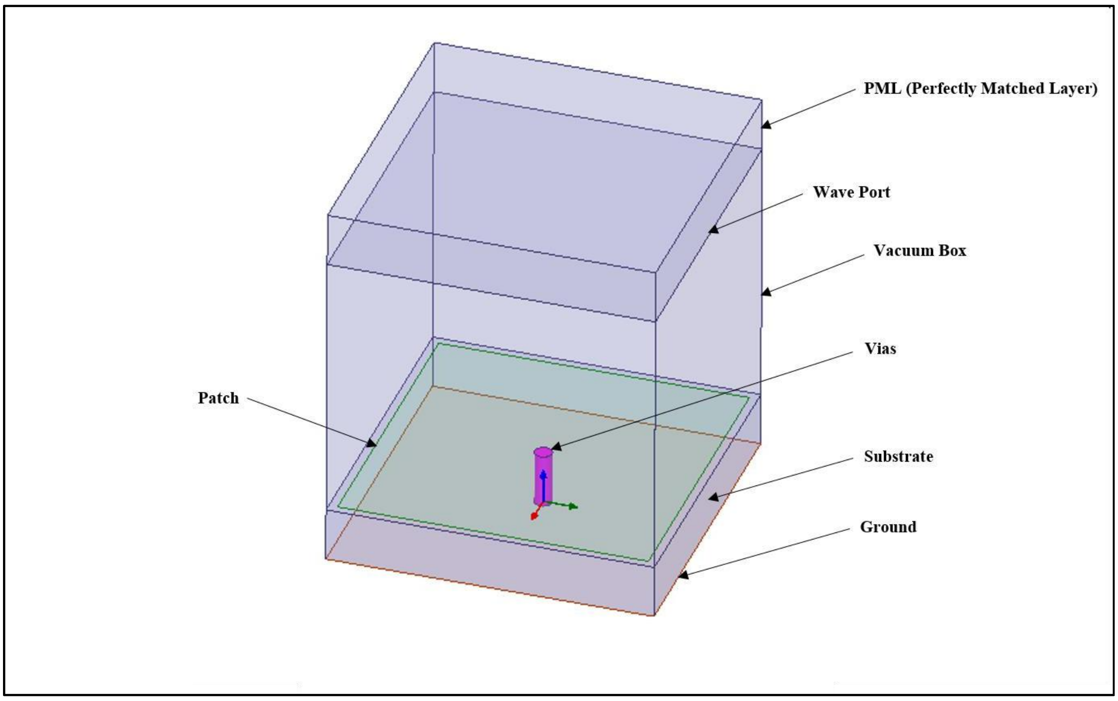


**Figure 6.13: Simulated EBG Unit Cell**

## Chapter 7

# HARDWARE IMPLEMENTATION

### 7.1 Fabrication Process

The fabrication of a typical patch antenna is based on the photo etching process. It utilizes a mask to selectively protect certain parts of a material that is exposed to ultraviolet radiation.

**1. Generate Mask**

First step is to print the simulated antenna's top and bottom layer structure onto the transparency film.

**2. Photo Exposure**

Second step is the Ultraviolet exposure process. It is done to transfer the image of the circuit pattern with a film in a UV exposure machine onto to the photo resist laminated board. The substrate is thoroughly cleaned using acetone and dried to remove the dust particles or impurities present on the substrate surface which introduce discontinuity in the etched pattern that alters the resonant frequency. Photo resist layer is uniformly applied over the substrate by controlled spinning. Bake the substrate and allow it to cool down. A negative photo resist film is laminated to the cleaned and dried substrate. The negative mask prepared in the first step is firmly placed on the photoresist laminated substrate. The masked and photoresist laminated substrate is exposed to the ultraviolet (UV) for few minutes. The UV exposed photoresist laminated substrate becomes hard and dark blue in color while unexposed photoresist remains light blue and dissolves in the developer solution. Sodium Carbonate is used as the developer.

**3. Etching**

Third step is the etching process using ferric chloride. Its purpose is to remove the unwanted copper area. This process is followed by the removal of the solution by water. The developed substrate is chemically etched by ferric chloride ($FeCl_3$) solution. The copper parts except the one underneath the hardened photoresist dissolves in $FeCl_3$. The etched substrate is rinsed in running water to remove any etchant and then dried. The hardened photoresist is removed using sodium hydroxide. The photo-lithographically fabricated microstrip slot antenna

**4. Soldering Probe**

The fourth step in the fabrication is soldering. A coaxial cable is used to excite using Sub-Miniaturized Version-A (SMA) connector. The inner conductor of the SMA port is connected to the patch through the substrate using silver paste while the inner conductor of the SMA is soldered to the copper patch.

**5. EBG Fabrication**

Fabrication of the EBG structures are difficult compared to the general microstrip antenna. In case of EBG ground plane, a separate substrate layer is taken and EBG patches are printed following the same steps as that of patch antenna. However, once the EBG patches are created, each EBG patch is drilled at the center using a 1 millimeter drill bit. These cavities are filled with conductive elements in order to connect the EBG patch to ground. A multi-meter is used to check if all EBG structures are connected to the ground.

**6. Antenna Measurements**

The measurement is done in order to investigate the performance of the fabricated antenna. The Return Loss, Smith Chart and VSWR are analyzed and investigated. From the return loss, we also can observe the impedance bandwidth. Our goal is to compare the simulated results with the actual results obtained after fabrication of the design. These measurements are performed using an instrument called as Vector Network Analyzer (VNA).

## 7.2 Fabricated Hardware

The antenna designs showcased in the software implementation chapter have been fabricated. The Circular Polarized Patch Antenna Array with EBG is shown in Figure 7.1 and Figure 7.2 respectively.

### 7.2.1 Fabricated Circular Polarized Patch Antenna Array with EBG

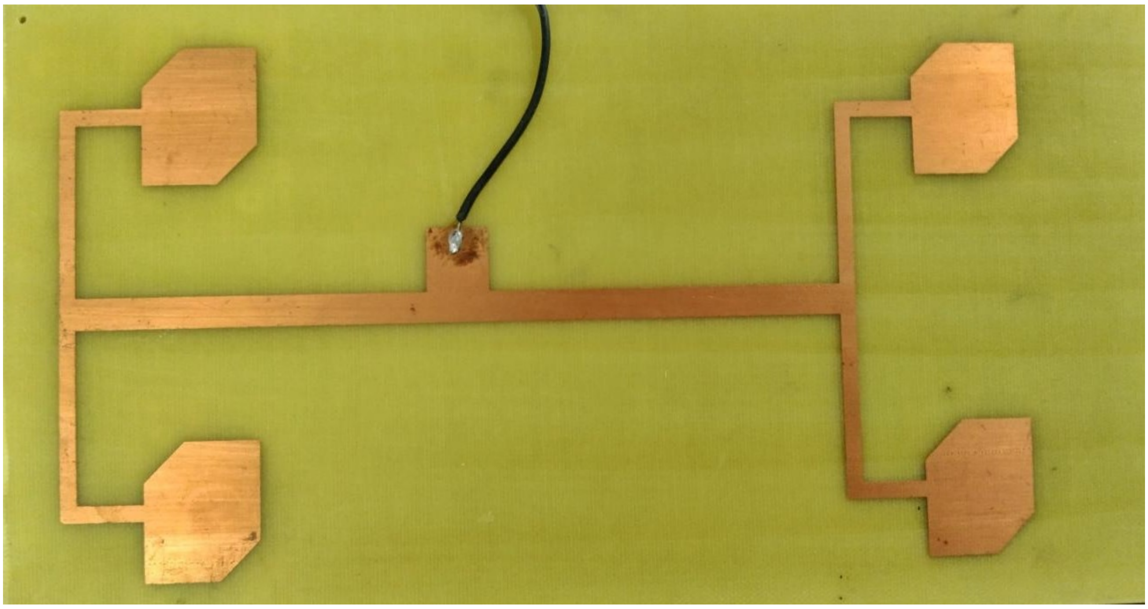

**Figure 7.1 : Circular Polarized Patch Antenna Array with EBG**

**Figure 7.2 : Circular Polarized Patch Antenna Array with EBG(Back view)**

# Chapter 8
# RESULTS

## 8.1 Simulation Results

The HFSS simulation tool was opted to analyze the results of our proposed antenna designs. Some of the major antenna parameters such as Return Loss, Gain, Radiation Pattern and Efficiency, Axial Ratio, Input Impedance and Directivity are measured.

### 8.1.1 Simulation of Circular Polarized Patch Antenna Array with EBG

**Dimensions**

The design of Circular Polarized Patch Antenna Array with EBG ground plane structures was based on Parametric Study. Parametric Analysis involves variation of dimensional variables of design in order to determine their effect on performance metrics. Effects of various variables of antenna were studied to derive at the result.

**Return Loss**

The Figure 8.1 depicts the Return Loss of Circular Polarized Patch Antenna Array with EBG ground plane. The antenna is designed to resonate at 2.4 GHz. However, due to truncation of edges and creating an array, a resonance is obtained. At 2.4 GHz exhibiting -37.42 dB return loss which is under acceptable level of losses. The impedance bandwidth is observed to be 180 MHz, which is suitable for wideband applications.

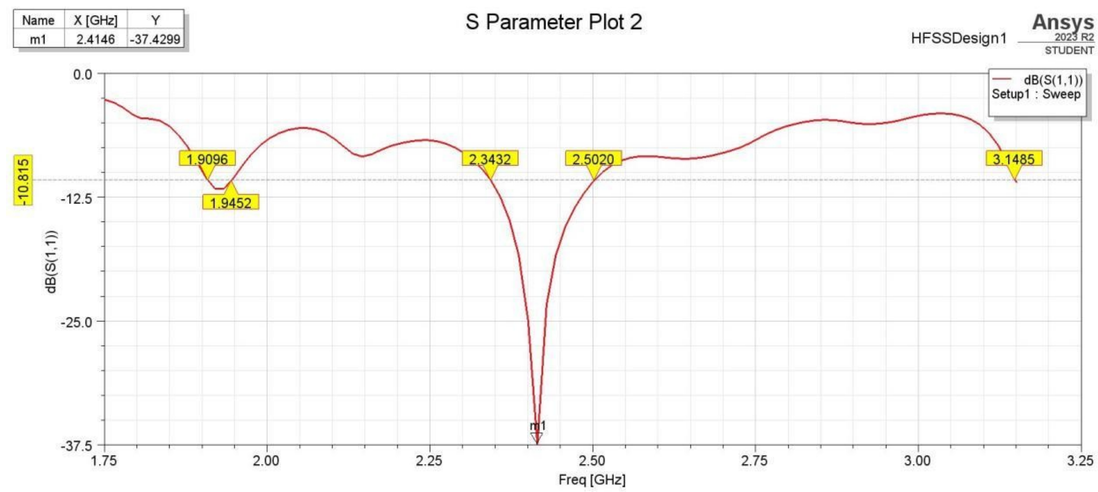


**Figure 8.1: Return Loss for Circular Polarized Patch Antenna Array with EBG**

### Voltage Standing Wave Ratio (VSWR)

The Figure 8.2 depicts the VSWR plot for antenna under consideration. It is observed that its value is 1.02 (absolute) at 2.4 GHz This value is close to 1, in the frequency range of operation of the antenna. This shows that a good impedance matching has been achieved and reflections minimized.

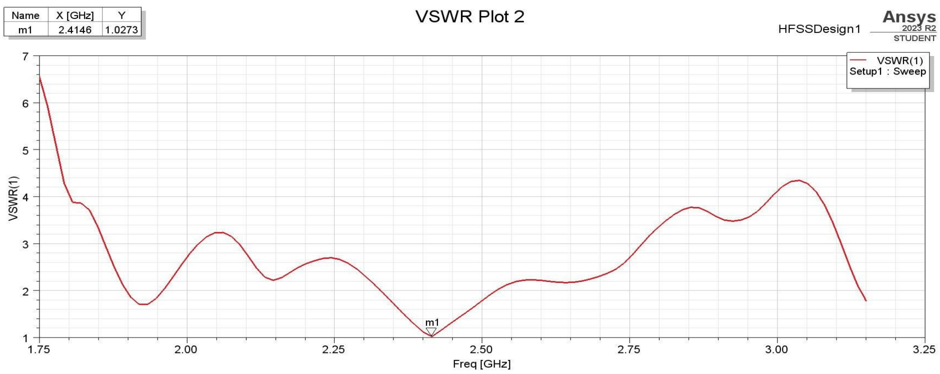


**Figure 8.2: VSWR for Circular Polarized Patch Antenna Array with EBG**

### Gain

The Figure 8.3 shows the three dimensional plot of gain for the antenna under consideration. It is observed that gain is 7.49 dB Also, it is observed that back lobe has been minimized.

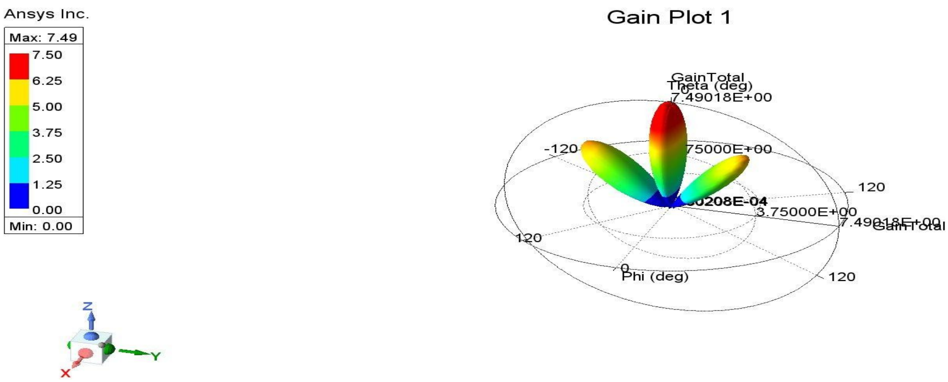


**Figure 8.3 : 3D Gain for Circular Polarized Patch Antenna Array with EBG**

## Axial Ratio

The Figure 8.4 depicts the Axial Ratio plot of the antenna under consideration, as a function of frequency. It is clearly observed that at the resonant frequency of 2.4 GHz, an axial ratio of 1.3 dB is observed, which is fairly close to 1 dB. This is considered to be an excellent value. The Figure depicts the polar plot of axial ratio.

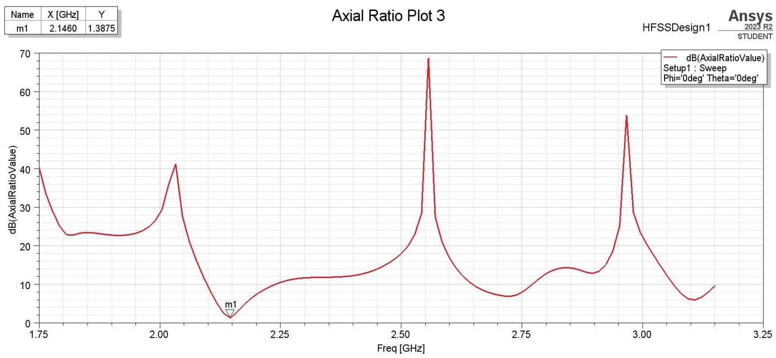


**Figure 8.4: Axial Ratio Plot for Circular Polarized Patch Antenna Array with EBG**

Axial ratio bandwidth plays a vital role in determining the operational characteristics of the antenna. This is because for the antenna to operate as circular polarized antenna, the axial ratio should be below 3dB level in the frequency range of operation.

## Electric Field Magnitude

The Figure 8.5 showcases the magnate of Electric Field Magnitude of the antenna under consideration. They are measure at 0 and 90 degrees phase. This is done in order to determine the strength of electric field as input phase. Also, it is used to determine the direction of circular polarization, which in our case is Left Hand Circul Polarization

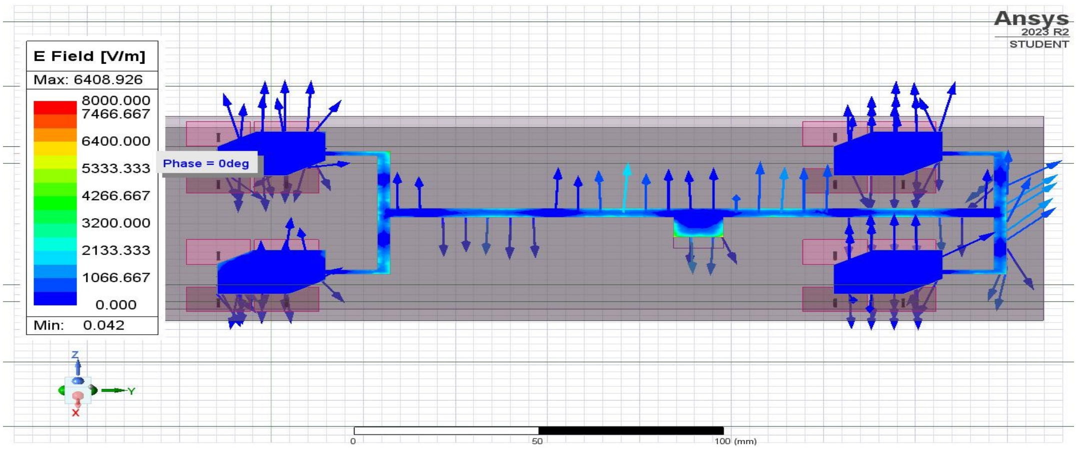


**Figure 8.5: Field Magnitude for Circular Polarized Patch Antenna Array with EBG**

### 8.1.2 Simulation of EBG cell

Impedance curve depicts the impedance as function of frequency of incident waves as shown in Figure 8.6. The EBG unit cell is designed at 2.4 GHz, where it exhibits the highest impedance. However, the high impedance is observed over wide frequency range centered at 2.4 GHz. This high impedance suggests that the surface waves are attenuated due to band stop nature.

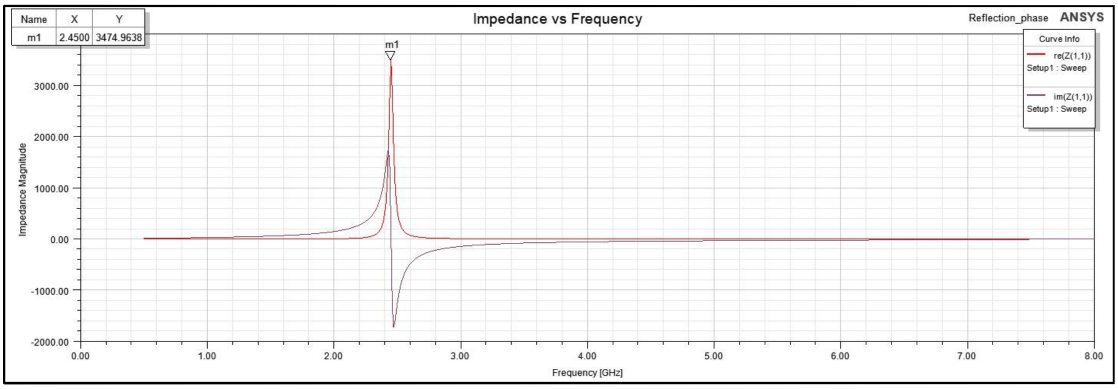


**Figure 8.6: Impedance Curve for EBG Unit Cell**

**Phase Reflection Curve**

The Figure 8.7 depicts the Phase Reflection Curve of the EBG unit cell. The backward radiated fields of the antenna are reflected back in phase which undergoes quasi-constructive interference with the forward radiation, which appears from -90 to +90 degrees phase. The bandwidth in this range is observed to be 390 MHz.

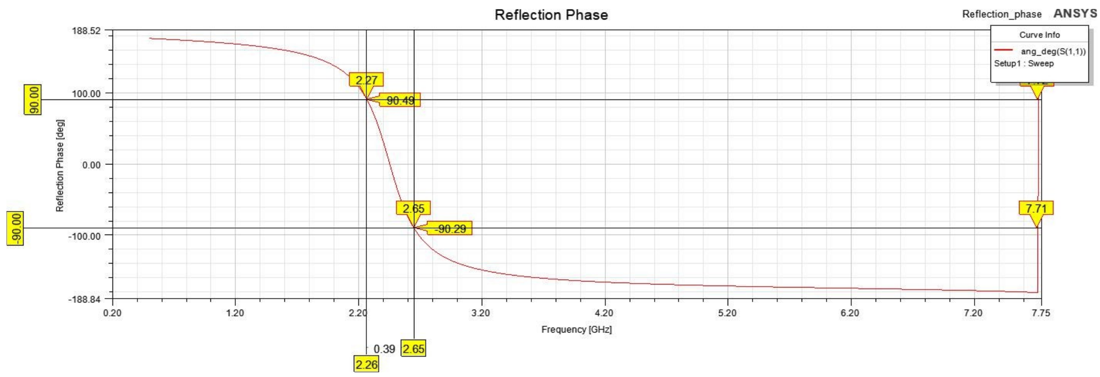


**Figure 8.7: Reflection Phase for EBG Unit Cell**

## 8.2 Fabrication Results

To verify the simulation results, antenna was fabricated and tested using an instrument called Vector Network Analyzer (VNA). Testing is necessary in order to ascertain the validity and accuracy of simulation. Simulations may not always provide the absolute perfect analysis because it involves major approximations

### 8.2.1 Fabricated Circular Polarized Patch Antenna Array with EBG

The Figure 8.8 depicts the Return Loss of fabricated version of Circular Polarized Patch Antenna Array with EBG, measured using Vector Network Analyzer (VNA). The Figure 8.9 and Figure 8.10 shows the VSWR and Smith Chart of the same antenna.

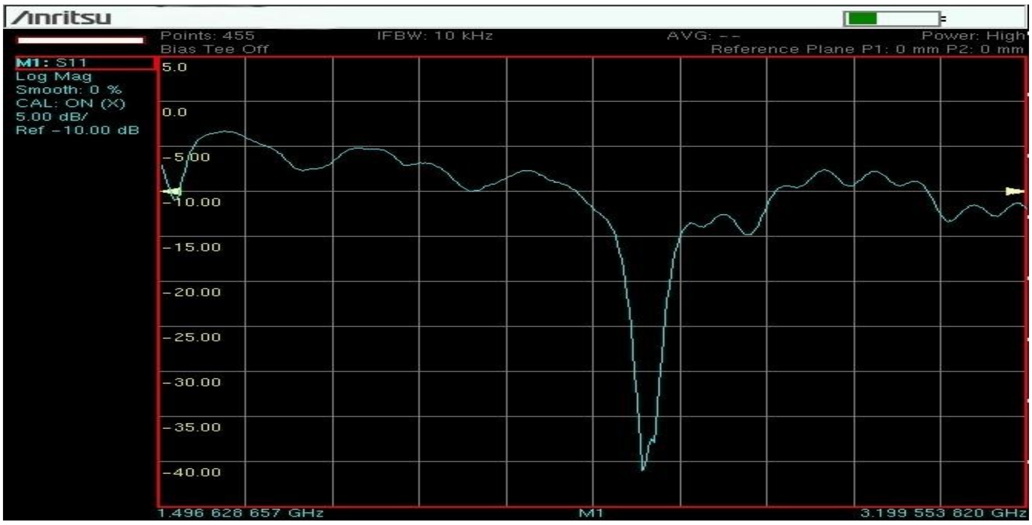


**Figure 8.8(a) : VNA Return Loss for Circular Polarized Patch Antenna Array with EBG**

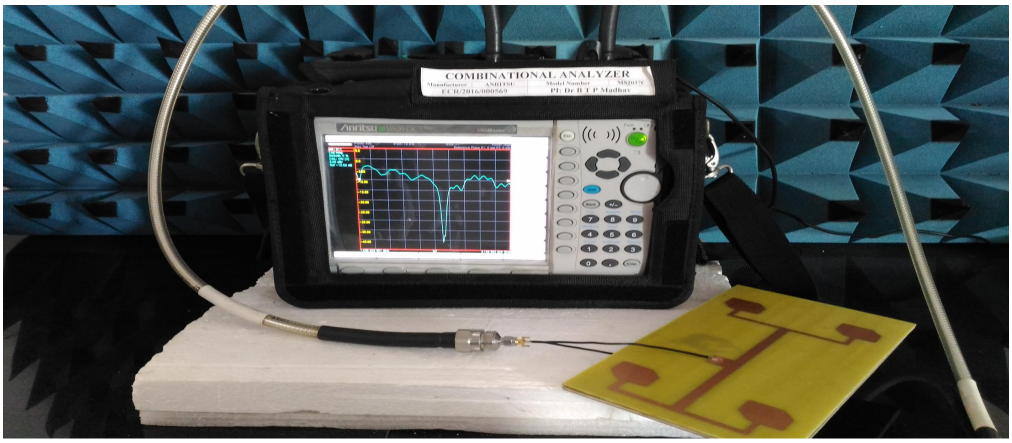


**Figure 8.8(b): VNA Return Loss for Circular Polarized Patch Antenna Array with EBG**

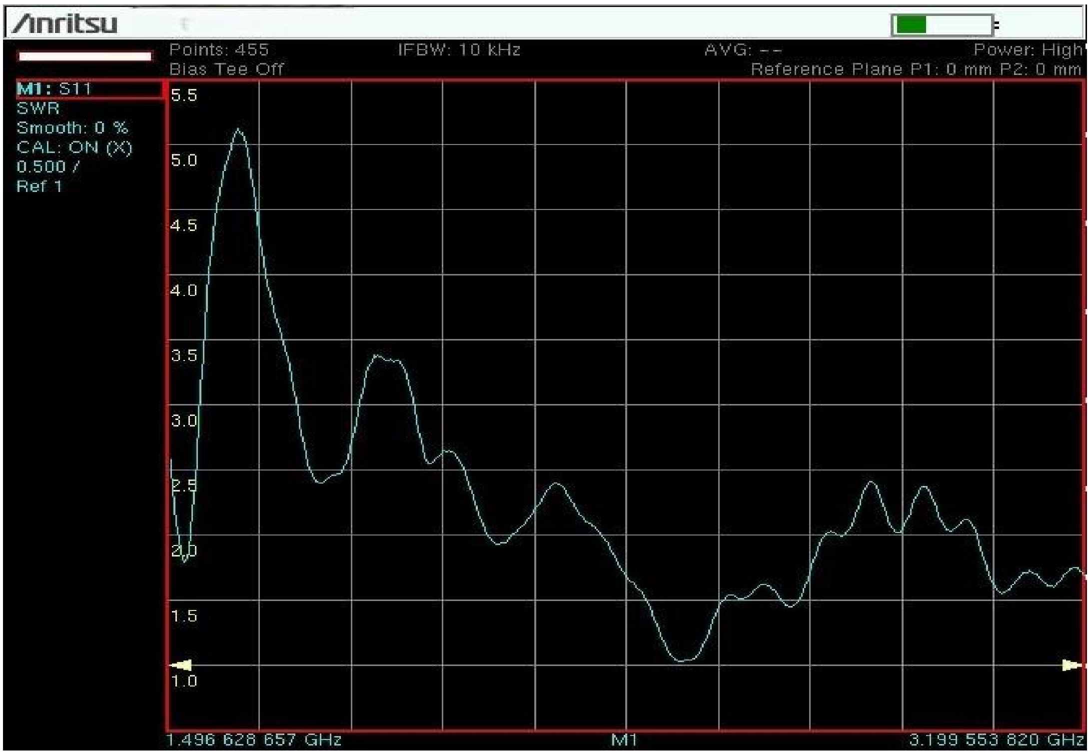


**Figure 8.9: VNA VSWR for Circular Polarized Patch Antenna Array with EBG**

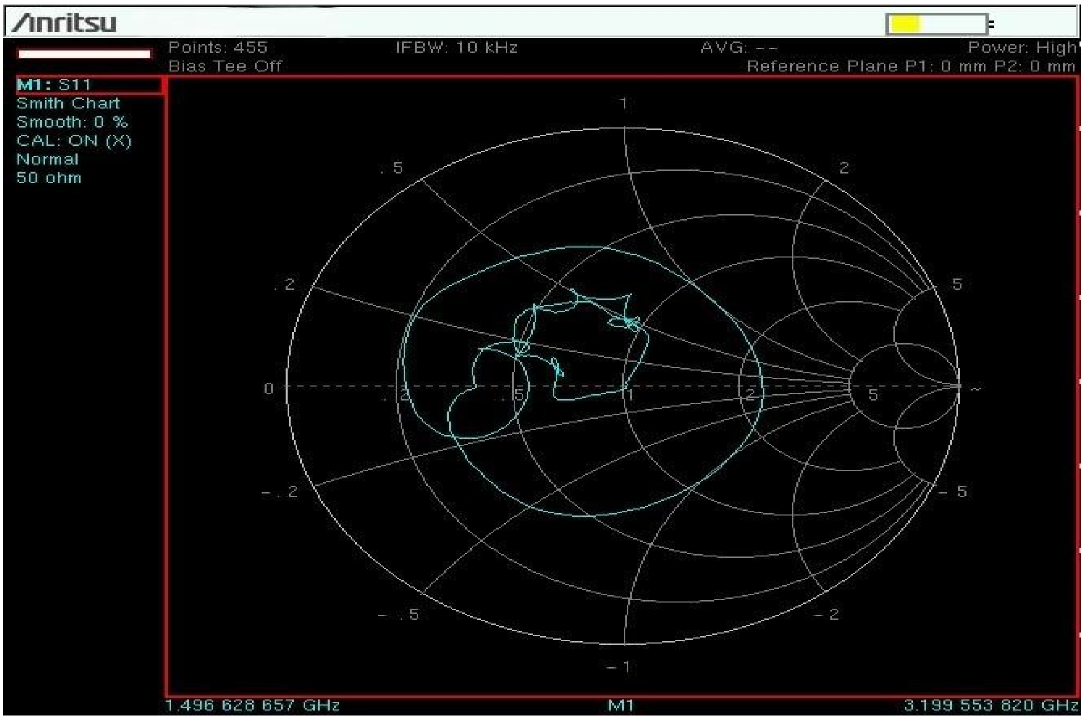


**Figure 8.10: VNA Smith Chart for Circular Polarized Patch Antenna Array with EBG**

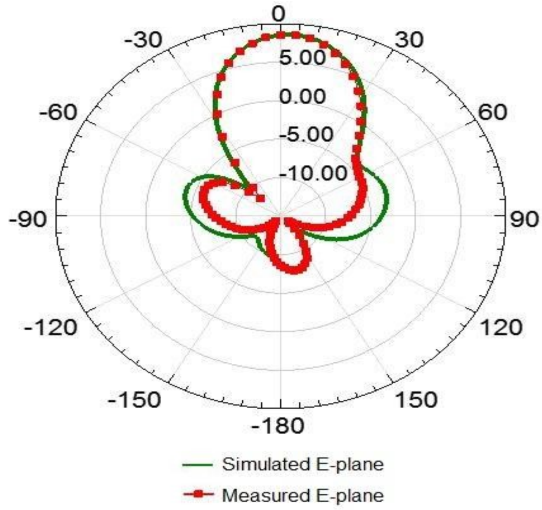


**Figure 8.11(a): Simulated and Measured Electric Plane for Circular Polarized Patch Antenna Array with EBG**

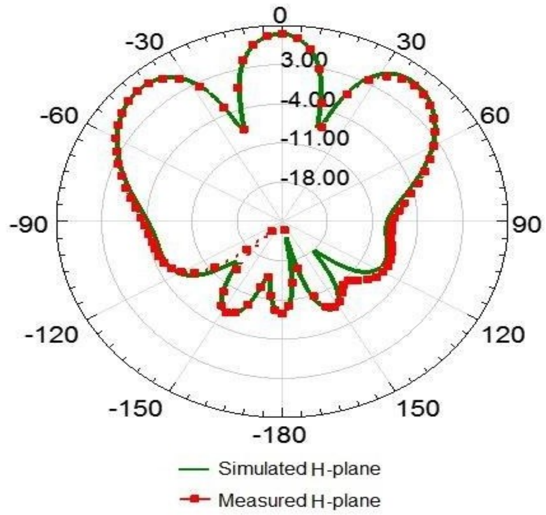


**Figure 8.11(b): Simulated and Measured Magnetic Plane for Circular Polarized Patch Antenna Array with EBG**

The above figures 8.11(a) & (b) indicate simulated and measured E plane and H plane for circular polarized patch antenna array using EBG where the antenna is resonating at 2.4 GHz.

## 8.3 Correlation of Simulated and Fabricated Results

It is important to correlate simulated results with fabricated results to ascertain the appropriateness of the design. Errors might be involved in simulated, because it itself is an approximation. The Table 8.1 depicts the comparison of Simulated and Fabricated results.

**Table 8.1: Comparison of Simulated and Fabricated Results**

| Parameter | Return Loss (dB) | VSWR (Abs) | Bandwidth (MHz) | Gain (dB) |
|---|---|---|---|---|
| Simulated | -37.42 | 1.02 | 180 | 7.49 |
| Measured | -39.9 | 1.03 | 210 | 8.02 |

# Chapter 9
# CONCLUSION

## 9.1 Conclusion

A square microstrip patch antenna array supporting circular polarization with Mushroom-type two dimensional EBG high impedance surface structures is presented with enhanced antenna performance. This antenna is titled as Circular Polarized Patch Antenna Array with EBG-high impedance surface. The antenna is designed to operate at license free ISM band, resonating at 2.4 GHz. To identify the enhancement in antenna performance due to EBG structure, an array of antennas are presented. These antennas were simulated using the HFSS tool. Parametric Study was executed in order to determine the effect of variation of dimensional parameters on the performance characteristics of proposed antennas. Excellent performance metric at desired specifications were achieved.

A detailed study along with literature survey was undertaken, to identify the best techniques to design the antennas. The major study included the comparison of substrate material and shape of the patch. A comprehensive comparison of various feeding techniques as well as circular polarization techniques were performed. FR-4 was opted as the substrate material because of its moderate electrical properties but most importantly for the ease of availability and cost associated. Square patch shape was chosen due to the good circular polarization characteristics caused by dimensional symmetry. Quarter Wavelength Feed was opted as the feeding technique because of the good impedance matching capabilities and non-destructive involvement in fabrication process. Truncation of two opposite edges of patch antenna was chosen as the technique for employing circular polarization due to its reduced complexity. Mushroom two-dimensional EBG was chosen because of moderate complexity as compared to other Photonic Band Gap structures. An array was created to support wider applications.

The Circular Polarized Patch Antenna Array with EBG surface on simulation, it was observed that due to the incorporation of EBG surface ,the bandwidth observed was 180MH and return loss was -37.42dB. An excellent gain from 7.49dB was observed due to the presence of EBG surface. Significant reduction in patch dimensions were also observed and radiation efficiency improved from 38% to 61% due to EBG layer. Also, the back lobe radiation was diminished. By the integration of

EBG surface with the patch antenna, the surface wave effect and mutual coupling is reduced, leading to enhancement of antenna performance metrics.

The Circular Polarized Patch Antenna Array with EBG high impedance surface was simulated in HFSS and then fabricated using photolithography process. Results were obtained by testing using Vector Network Analyzer. The EBG surface was placed below the patch, on the substrate. The EBG surface is connected to ground plane with vias. This fabricated antenna during testing using VNA was observed to be resonant at 2.4 GHz. It delivered an excellent impedance bandwidth of 180 MHz. At resonance, the return loss was observed to be -37.2 dB which is well above the operational requirements. Remarkable VSWR values for the frequency range of operation were achieved, which were found to be close to 1 (absolute) and 1.02 at resonance. The dimensions of the patch was significantly reduced, when compared to simple patch, which indicates miniaturization. The values of fabricated antenna extracted from VNA was found almost identical to the simulated values. By this, we conclude that the Mushroom EBG surface exhibits a great deal of potential, because of the enhancement in antenna performance metrics and ease of production and creation of array supports a wider

### 9.2Future Scope

Antennas with enhanced performance is highly desirable in applications ranging from commercial to military. Further, the EBG can be designed and fabricated for a conformal substrate to achieve an enhanced conformal antenna. Also, EBG layers without vias can be explored. They are often called as Unipolar EBG. They reduce the complexity of fabrication due to the elimination of stubs which require drilling and material deposition. Beam steering can be achieved by implementing non-uniform antenna array. The antenna can be further miniaturized by introducing slots within the patch in order to achieve circular polarization with even smaller patch dimensions. Further modifications and perturbations can be introduced to further increase the bandwidth of operation of the antenna to form ultra-wide band antennas. The antenna designs can further venture into the scope of Meta-Materials which opens up a whole new domain of concepts and plethora of applications.

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## APPENDIX-I

# SIMULATION PARAMETERS

**Table 9.1: Dimensions of Antenna Design**

| S.No. | Parameters | Value |
|---|---|---|
| 1. | Patch Length | 31.5 mm |
| 2. | Patch Width | 29.5 mm |
| 3. | Substrate Height | 4.72 mm |
| 4. | Height of EBG Ground Plane | 4.72 mm |
| 5. | Substrate Length | 53 mm |
| 6. | Substrate Width | 102.3 mm |
| 7. | Edge Feed Width | 3.59 mm |
| 8. | Edge Feed Length | 13.95 mm |
| 9. | Feed Width | 6.75 mm |
| 10. | Truncation Size | 11 mm |
| 11. | EBG Inter Element Spacing (Gap) | 1 mm |
| 12. | EBG Mushroom Width | 17.5 mm |

## APPENDIX-II

# DATASHEET

**Table 9.2: Datasheet for FR-4 Substrate**

| Substrate Parameter | Substrate Parameter |
|---|---|
| Dielectric constant | 4.4 |
| Substrate thickness | 1.6 mm |
| Loss tangent | 0.019 |
| Conductor (copper) thickness | 0.035 mm |